

EasyVisa

Machine Learning Project - 2

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1. Problem statement

1.1 Context

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permit foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

1.2 Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having

a higher chance of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You, as a data scientist at EasyVisa, have to analyze the data provided and, with the help of a classification model:

Facilitate the process of visa approvals. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

1.3 Data Description

case_id: ID of each visa application

continent: Information of continent the employee

education_of_employee: Information of education of the employee

has_job_experience: Does the employee have any job experience? Y= Yes;
N = No

requires_job_training: Does the employee require any job training? Y = Yes;
N = No

no_of_employees: Number of employees in the employer's company

yr_of_estab: Year in which the employer's company was established

region_of_employment: Information of foreign worker's intended region of employment in the US.

prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.

unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.

full_time_position: Is the position of work full-time? Y = Full-Time Position;
N = Part-Time Position

case_status: Flag indicating if the Visa was certified or denied

2. Import the libraries and functions:

- We are going to import NumPy and Pandas to help with reading and manipulating data.
- We are going to import Seaborn and Matplotlib to help with our data visualization.
- We are going to use 'train_test_split' function from our sklearn.model_selection library to split the data into train and test.
- We use 'f1-score', 'accuracy_score', 'recall_score', 'precision_score' and confusion_matrix from our sklearn.metrics library to check model performance.
- We import SimpleImputer to impute missing values.
- From Imblearn we import SMOTE and RandomUnderSampler.
- We import RandomizedSearchCV to search through hyperparameteres.
- We import DecisionTreeClassifier from sklearn.tree to build decision tree.
- We import AdaBoostClassifier, GradientBoostingClassifier, RandomForestClassifier, BaggingClassifier from sklearn.ensemble to build ensemble models.

3. Load the dataset:

We load the dataset using 'pd.read_csv' function and by giving the path location of our dataset.

4. Data Overview

This gives us a basic and important information of our dataset. The initial steps to understand a dataset is to:

- Observe the first few rows of the dataset to check whether the dataset is loaded properly or not.
- Get information about the rows and columns in the dataset.
- Find out the data types and non-null value counts of the column .

- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

4.1 Displaying first five rows of the dataset

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status
0	EZYV01	Asia	High School	N	N	14513	2007	West	592.2029	Hour	Y	Denied
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.6500	Year	Y	Certified
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.8600	Year	Y	Denied
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.0300	Year	Y	Denied
4	EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.3900	Year	Y	Certified

Table 4.1 - First five rows of dataset

4.2 Checking the shape of the dataset

(25480, 12)

Table 4.2 – Shape of dataset

There are 25480 rows and columns.

4.3 Checking the data types of the columns for the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   case_id                               25480 non-null  object
1   continent                             25480 non-null  object
2   education_of_employee                 25480 non-null  object
3   has_job_experience                     25480 non-null  object
4   requires_job_training                  25480 non-null  object
5   no_of_employees                       25480 non-null  int64
6   yr_of_estab                           25480 non-null  int64
7   region_of_employment                  25480 non-null  object
8   prevailing_wage                       25480 non-null  float64
9   unit_of_wage                          25480 non-null  object
10  full_time_position                     25480 non-null  object
11  case_status                           25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Table 4.3 – Datatype of the columns

- There are three numeric variables(float and int) and nine object type variables.
- The target variable is 'case_status' which is of object type.

4.4 Statistical summary of the dataset

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
case_id	25480	25480	EZYV25480	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.0	NaN	NaN	NaN	5667.04321	22877.928848	-26.0	1022.0	2109.0	3504.0	602069.0
yr_of_estab	25480.0	NaN	NaN	NaN	1979.409929	42.366929	1800.0	1976.0	1997.0	2005.0	2016.0
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.0	NaN	NaN	NaN	74455.814592	52815.942327	2.1367	34015.48	70308.21	107735.5125	319210.27
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Table 4.4 – Statistical summary of the dataset

- There are employees from 6 continent and most of the employees are from Asia appearing 16861 times.
- Most of the employees have Bachelor's degree, appearing 10234 times.
- 14802 employees have job experiences.
- Most of the employees do not require job training(22525).
- The average number of employees is 2109. The minimum number of employees is -26, which is invalid and need to be checked. The maximum number of employees is 602069. There is a huge difference in 75% quantile and maximum which suggest that there may be outliers in this variable.
- The oldest company establishment is in 1800 and the newest company establishment is in the year 2016.

- There are total of 5 regions where employees are intended for employment. Most employee are intended to be employed in Northeast region(7195).
- The average of prevailing wage is 70308.21. The minimum and maximum wages are 74455.8 and 319210.27.
- There are four unit of wages, from which yearly wages appear 22962 times.
- Most employees job position is for full time(22773).
- Visa has been certified for 17018 employees.

5. Exploratory Data Analysis

5.1 Univariate Analysis

5.1.1 Distribution of 'no_of_employees'

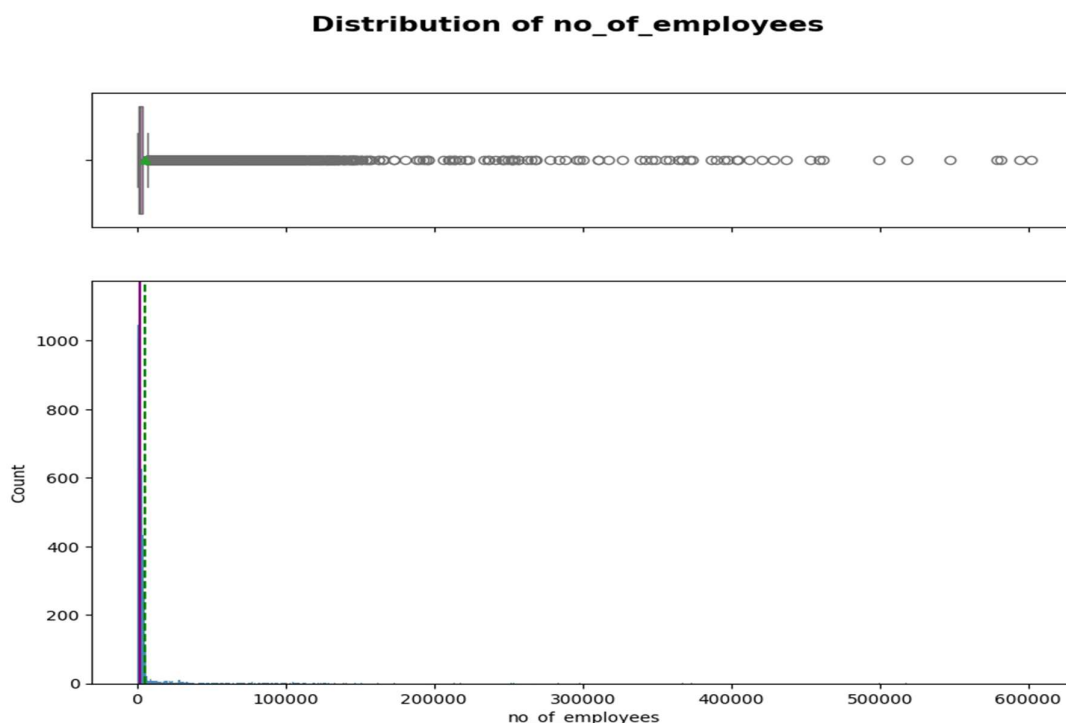


Fig 5.1.1 - Distribution of no_of_employees

- No of employee's distribution is heavily right skewed.

- There are some extreme outliers which implies that some companies have exceptionally large employee counts.
- Most observations are concentrated at low employee counts.

5.1.2 Distribution of 'yr_of_estab'

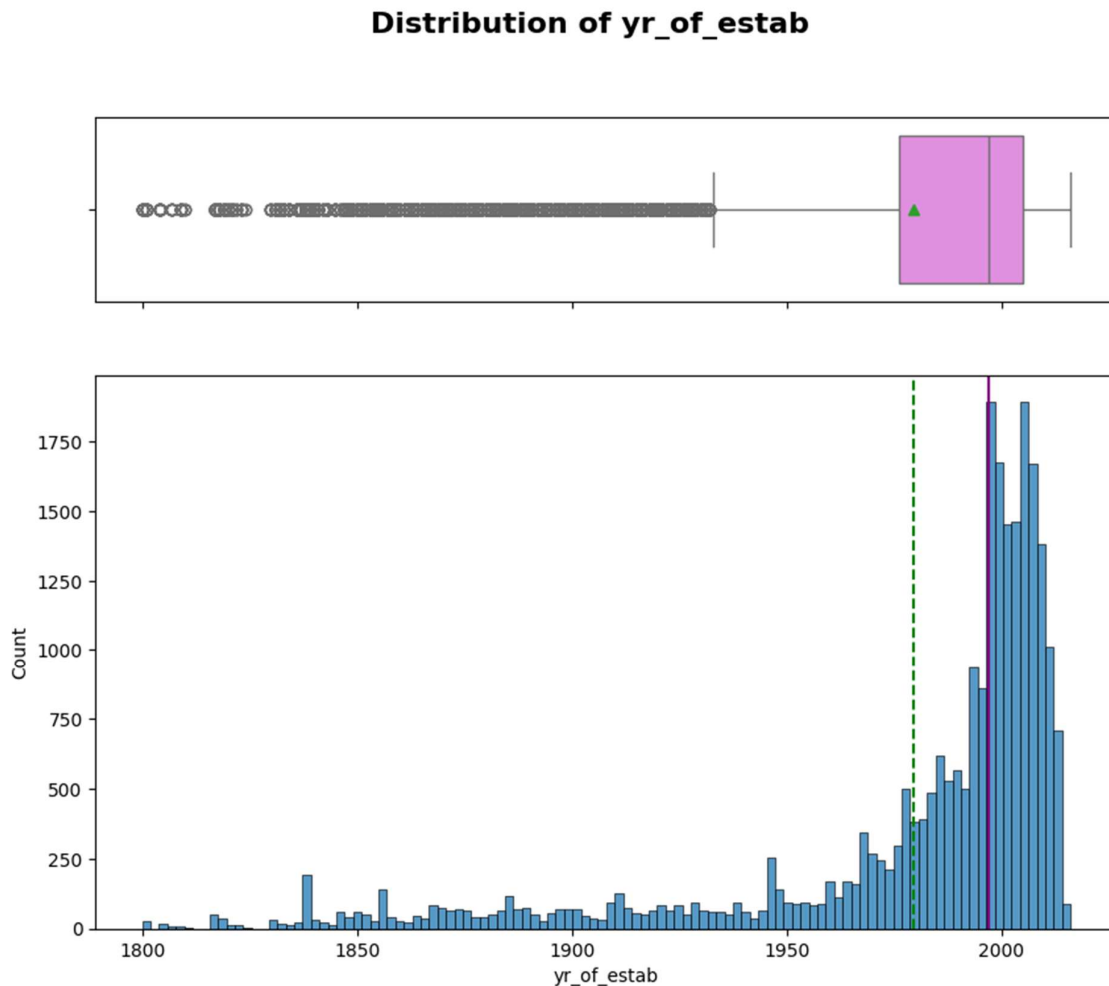


Fig 5.1.2 - Distribution of yr_of_estab

- The distribution of year of establishment is heavily left skewed.
- Most of the companies were established in the year 2000.
- There are some low outliers which implies company establishment around the year 1800.

5.1.3 Distribution of 'prevailing_wage'

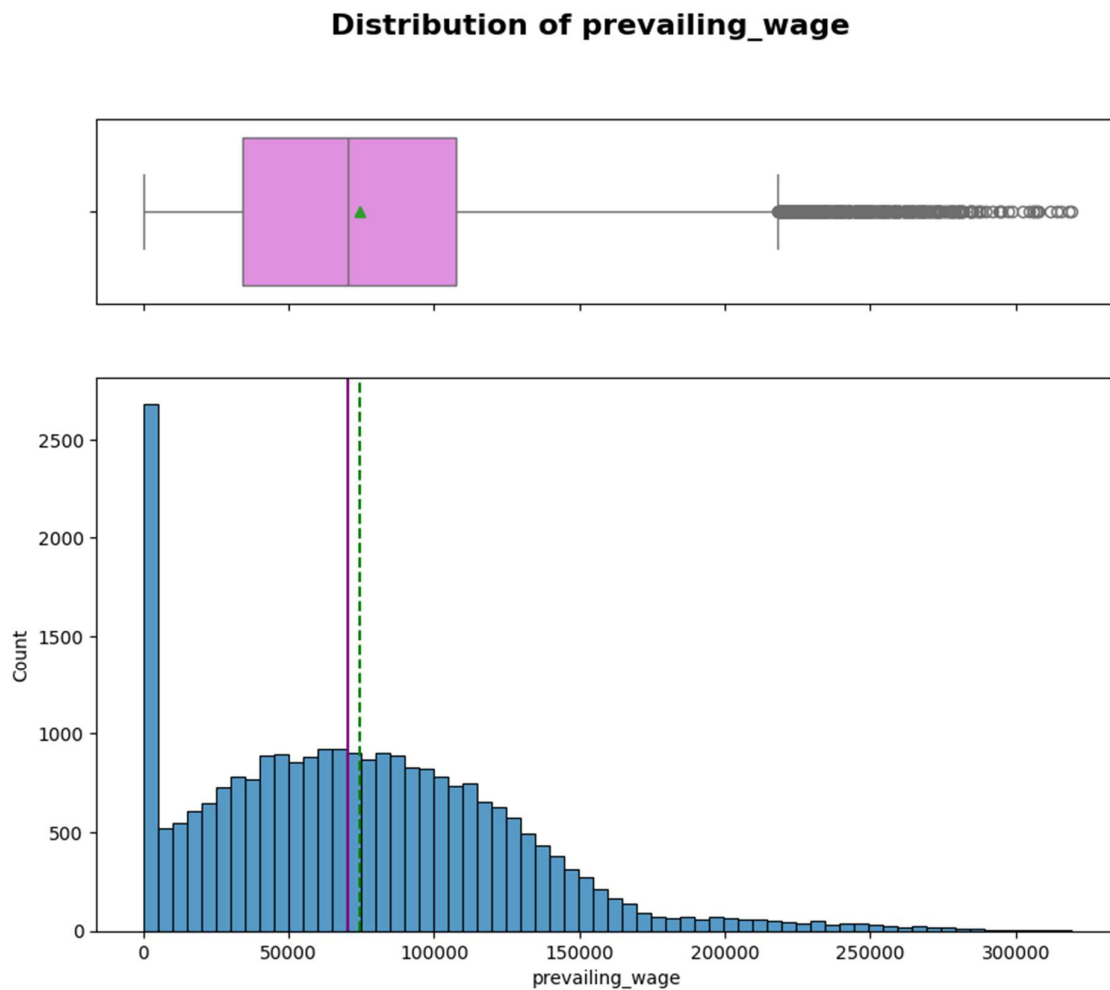


Fig 5.1.3 - Distribution of prevailing_wage

- The distribution of prevailing wage is right skewed.
- Most wages are concentrated in the lower-to-mid salary range.
- Some high value wages are shown as outliers.

5.1.4 Distribution of 'continent'

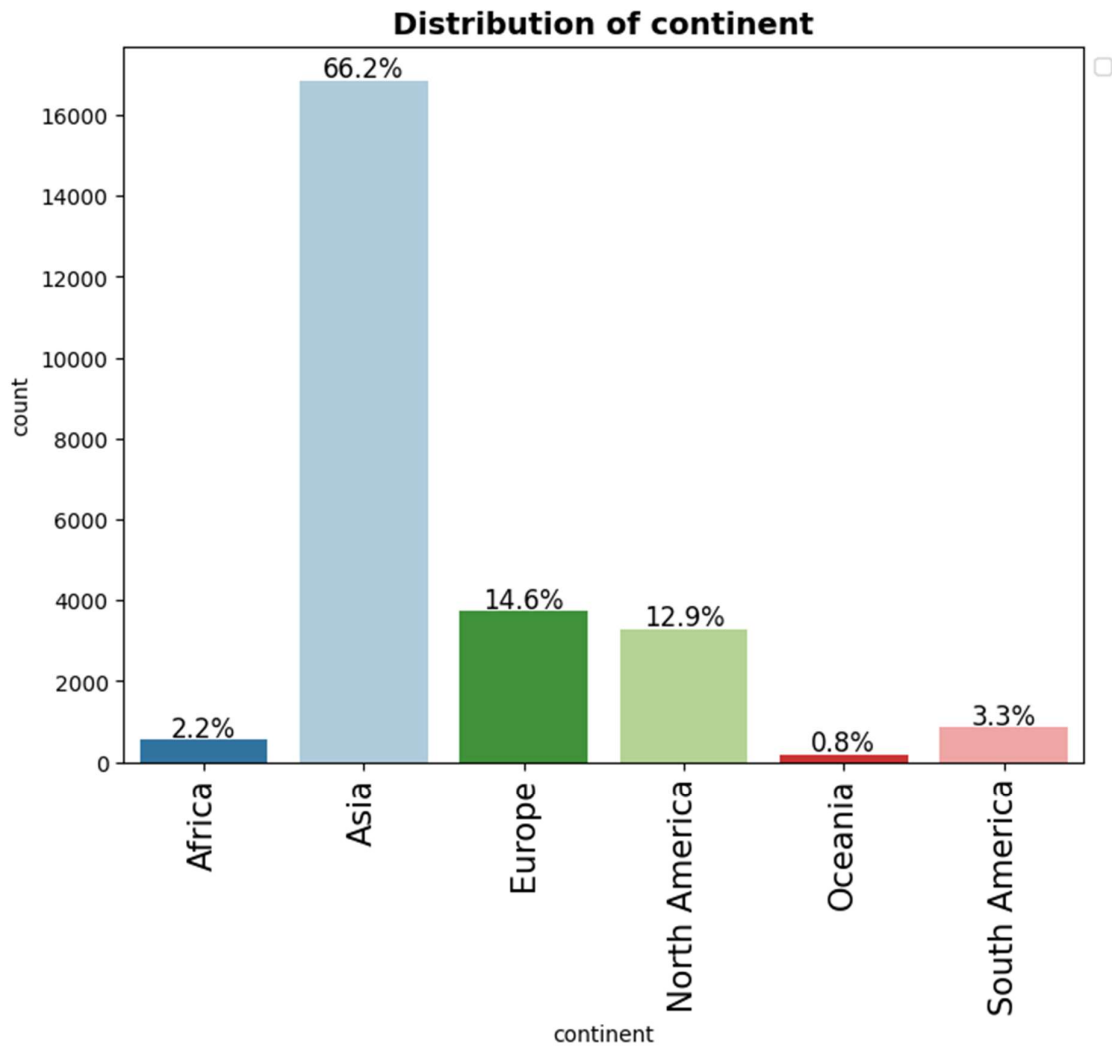


Fig 5.1.4 - Distribution of continent

- Asia has the highest count with 66.2%.
- Europe and North America form a secondary group with moderate representation.
- The distribution shows class imbalance across continents.

5.1.5 Distribution of 'education_of_employee'

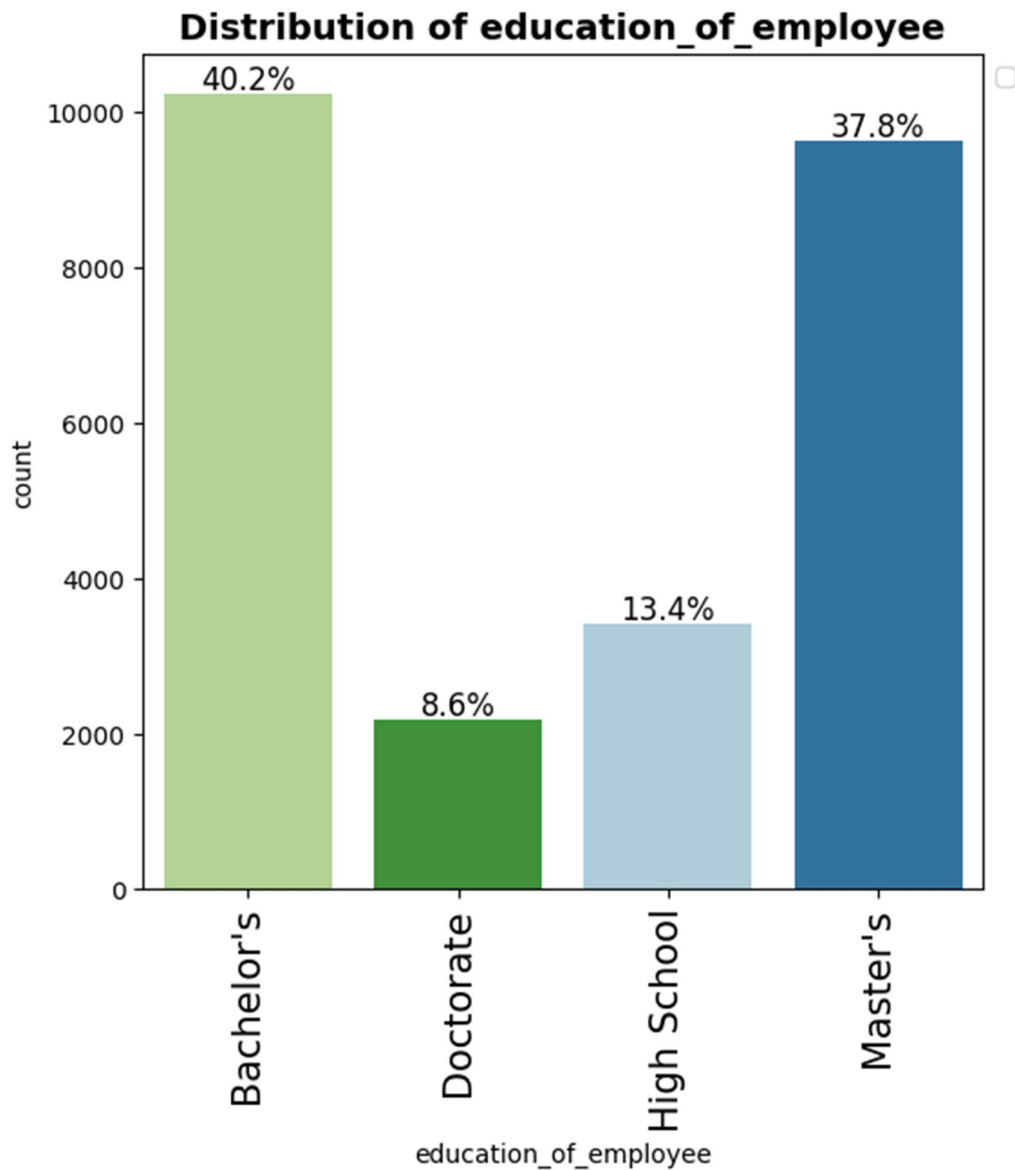


Fig 5.1.5 - Distribution of education_of_employee

- 40.2% of the employee have bachelor's degree.
- Around 37.8% of the employees have Master's degree.
- Employee who has completed high school is 13.4%.
- Doctorate employees are least at 8.6%.

5.1.6 Distribution of 'has_job_experience'

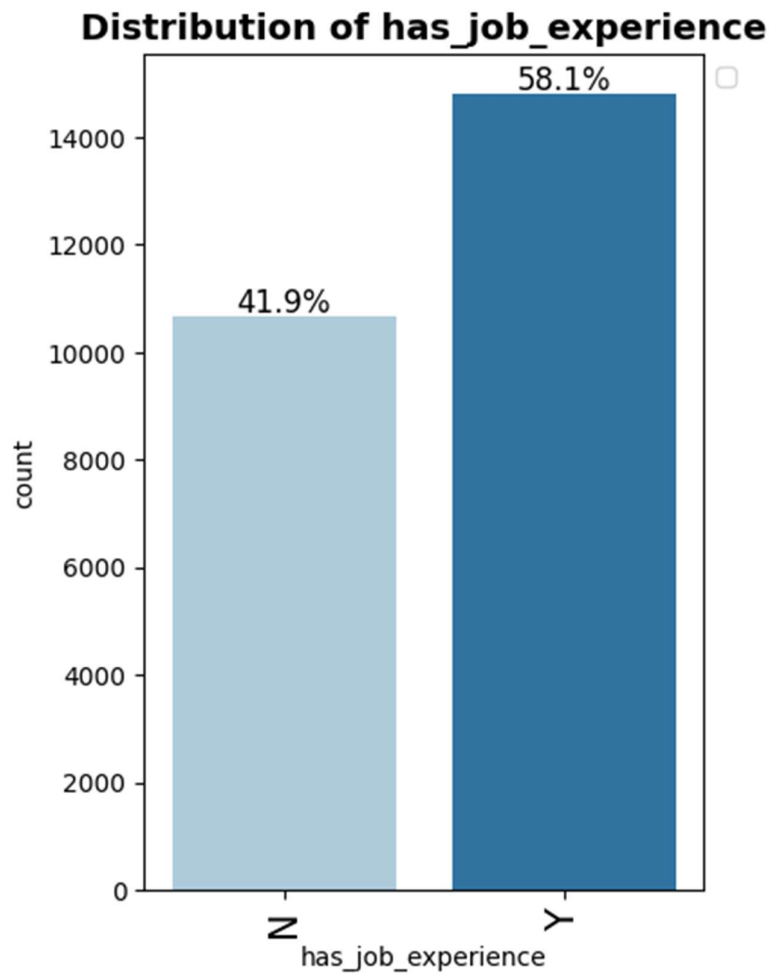


Fig 5.1.6 - Distribution of 'has_job_experience'

- 58.1% employees have job experience while 41.9% employee do not have job experience.

5.1.7 Distribution of 'requires_job_training'

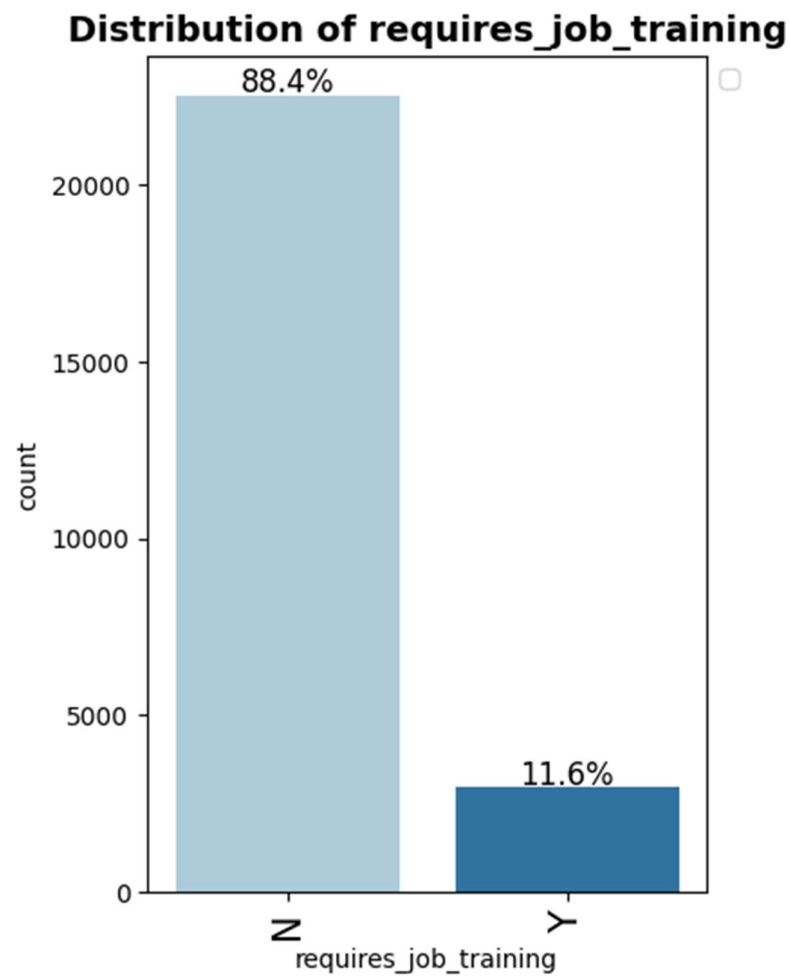


Fig 5.1.7 - Distribution of requires_job_training

- 11.6% of the employees required job training.
- 88.4% of the employees do not require job training.

5.1.8 Distribution of 'region_of_employment'

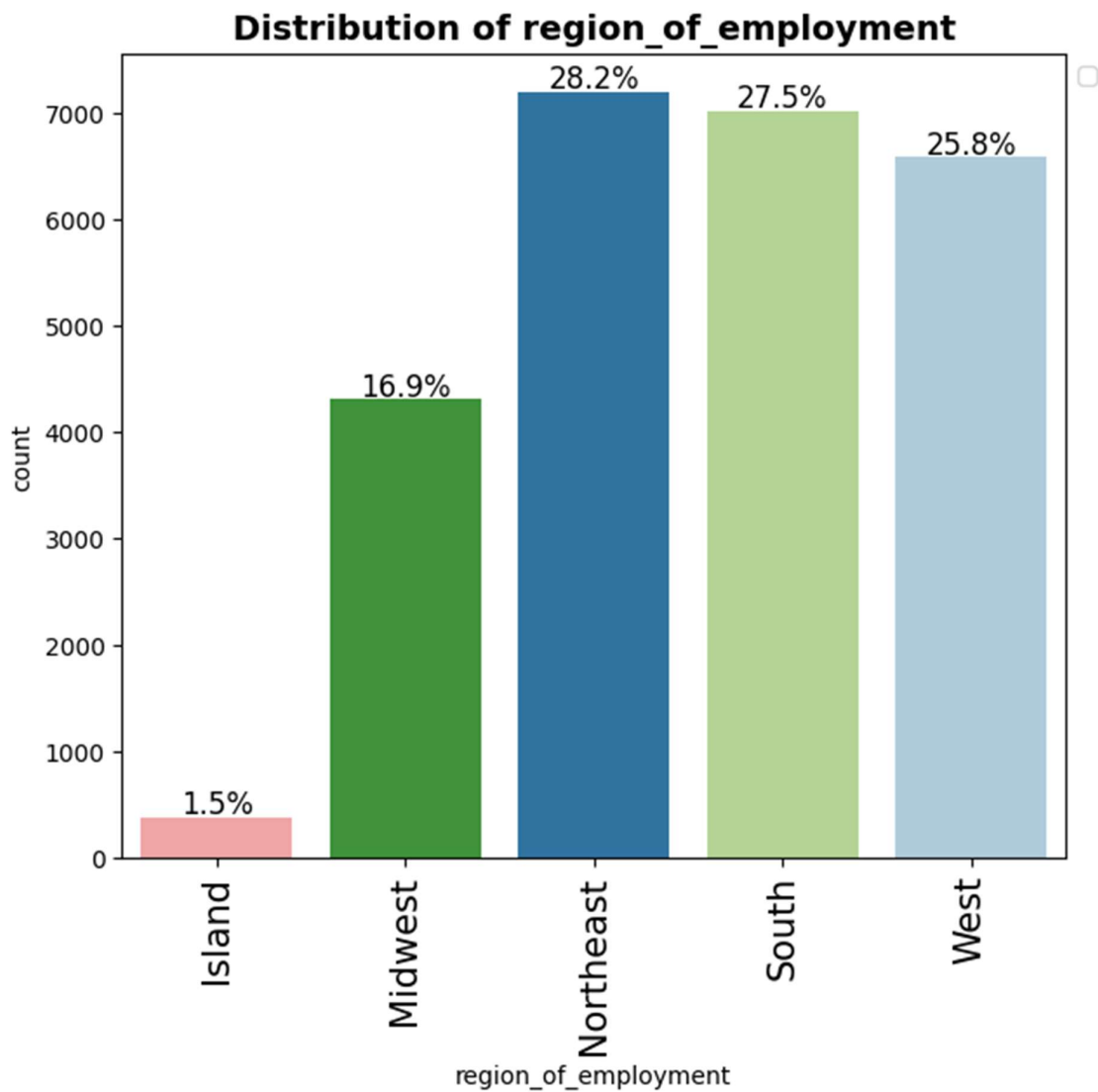


Fig 5.1.8 - Distribution of region_of_employment

- Most of the employees are intended to work in Northeast at 28.2%, followed by south at 27.2% and west region by 25.8%.
- 16.9% of the employees are intended to work in Midwest region.
- Only 1.5% of the employees are intended to work in Island

5.1.9 Distribution of 'unit of wages'

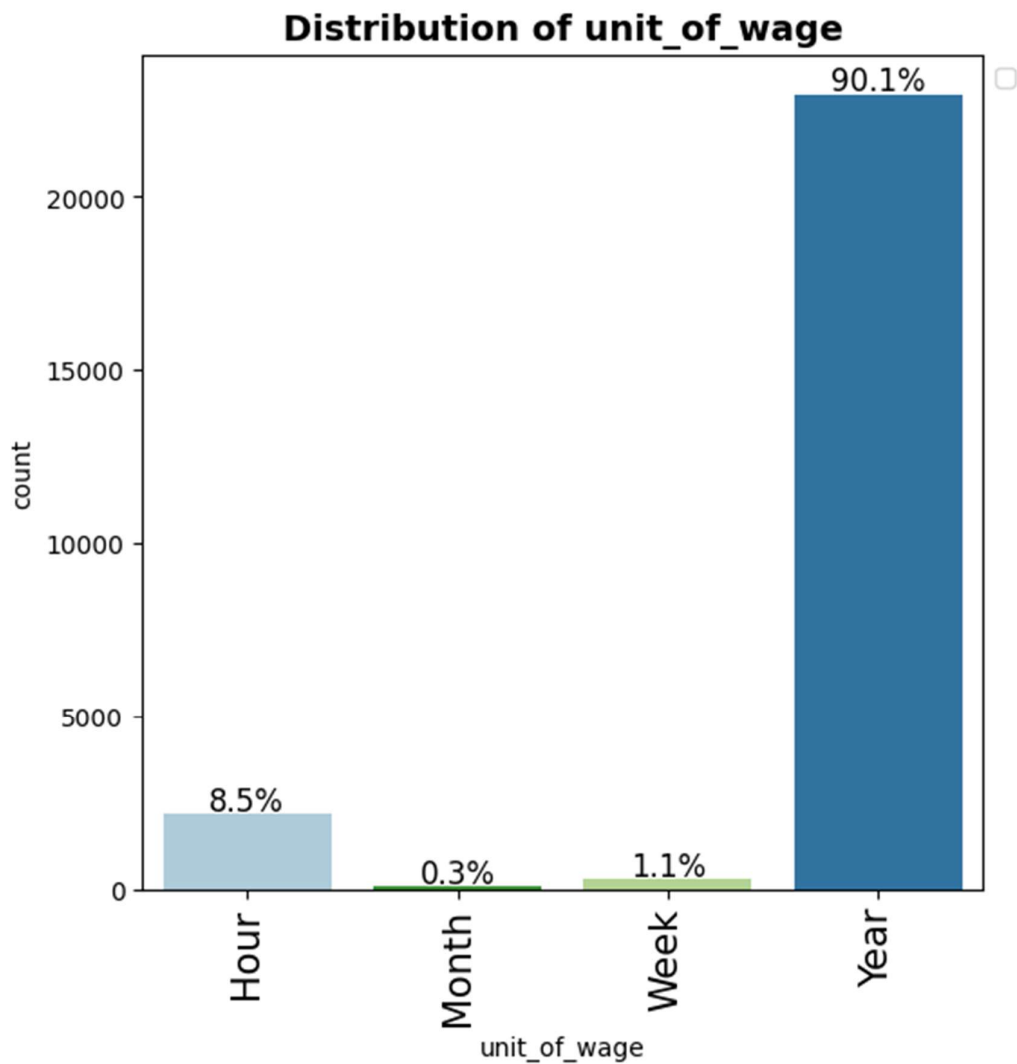


Fig 5.1.9 - Distribution of unit_of_wages

- The vast majority of wages are reported on a yearly basis (~90%), making it the dominant unit.
- Hourly wages account for 8.5%.
- Monthly and weekly wages account small for 0.3% and 1.1%.

5.1.10 Distribution of 'full_time_position'

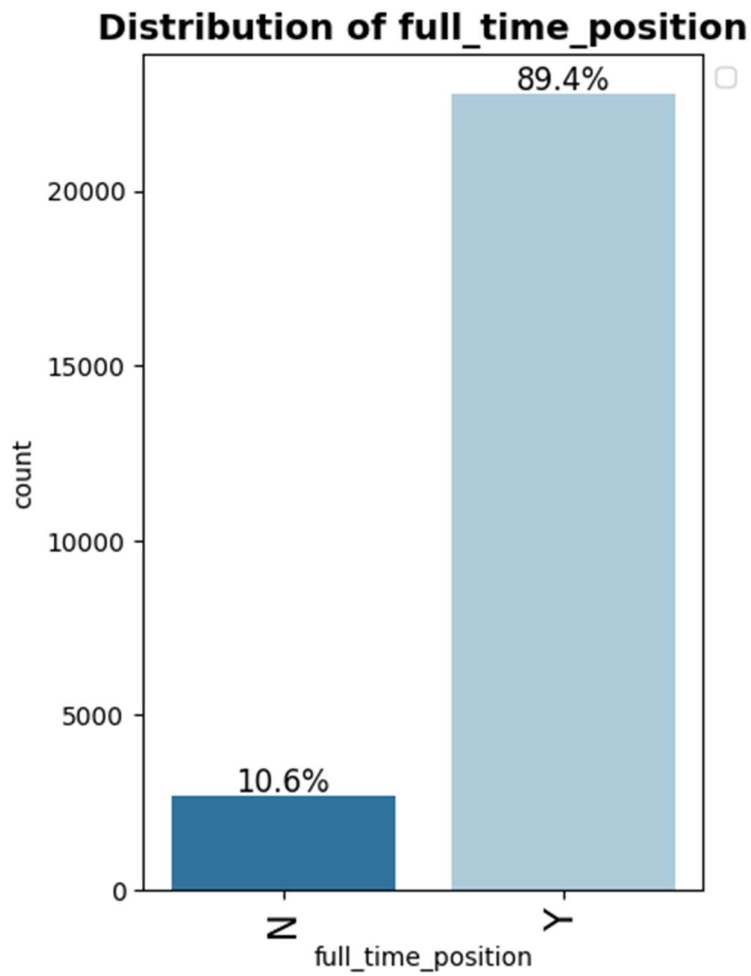


Fig 5.1.10 - Distribution of 'full_time_position'

- 89.4% of the employees are positioned for full time work.
- 10.6% of the employees are positioned for part-time.

5.1.11 Distribution of 'case_status'

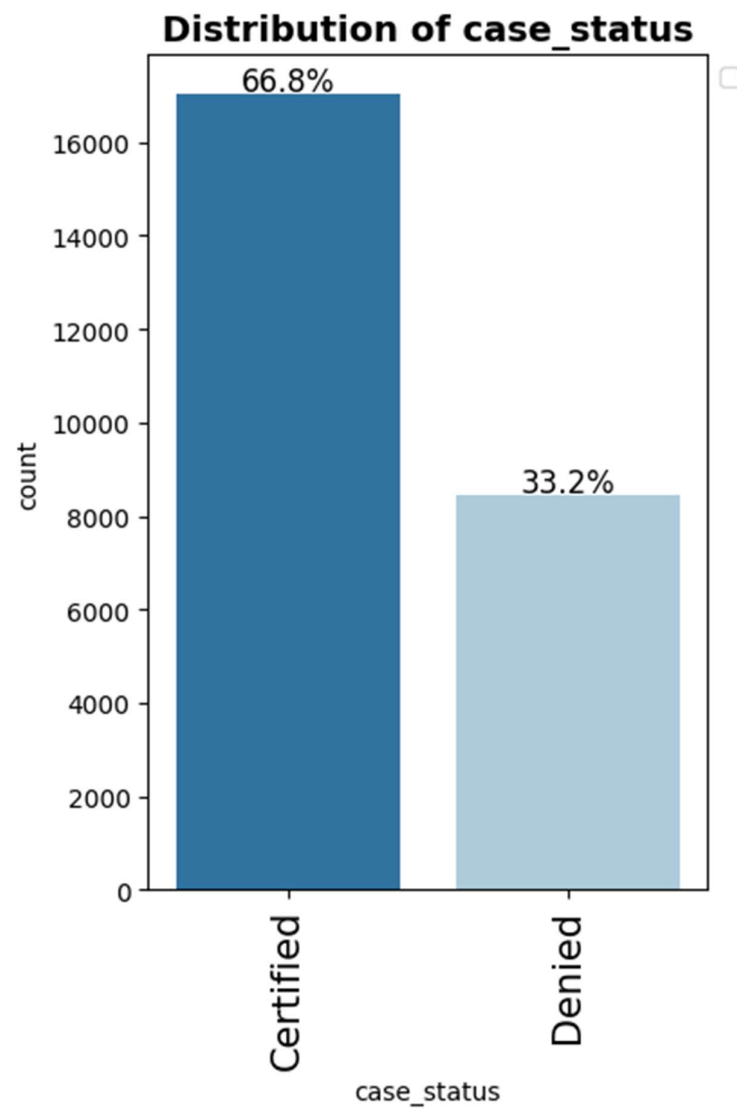


Fig 5.1.11 - Distribution of 'case_status'

- 66.8% of the employees were certified with visa.
- 33.2% of the employees were denied visa.

5.2 Bivariate analysis

5.2.1 Pairplot

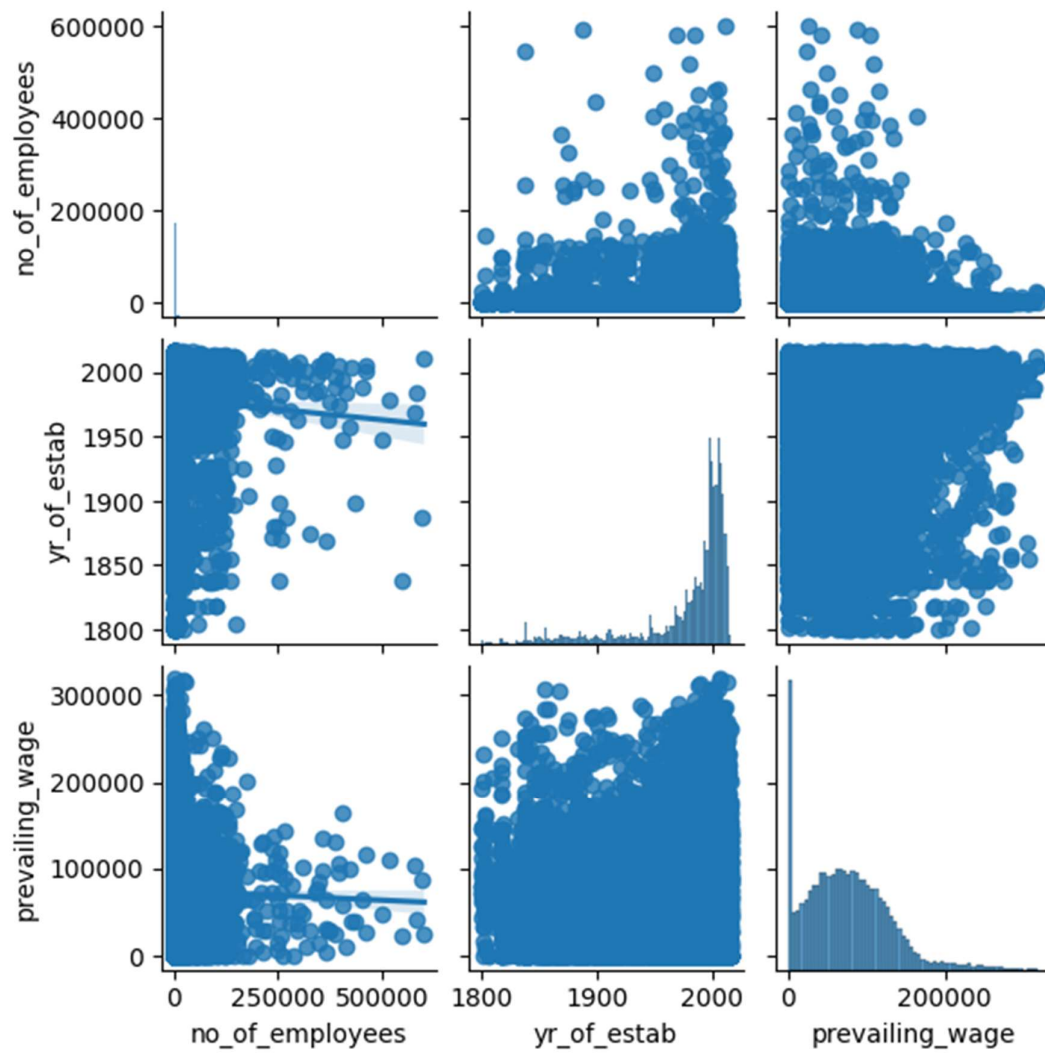


Fig 5.2.1 - Pairplot

5.2.2 Heatmap

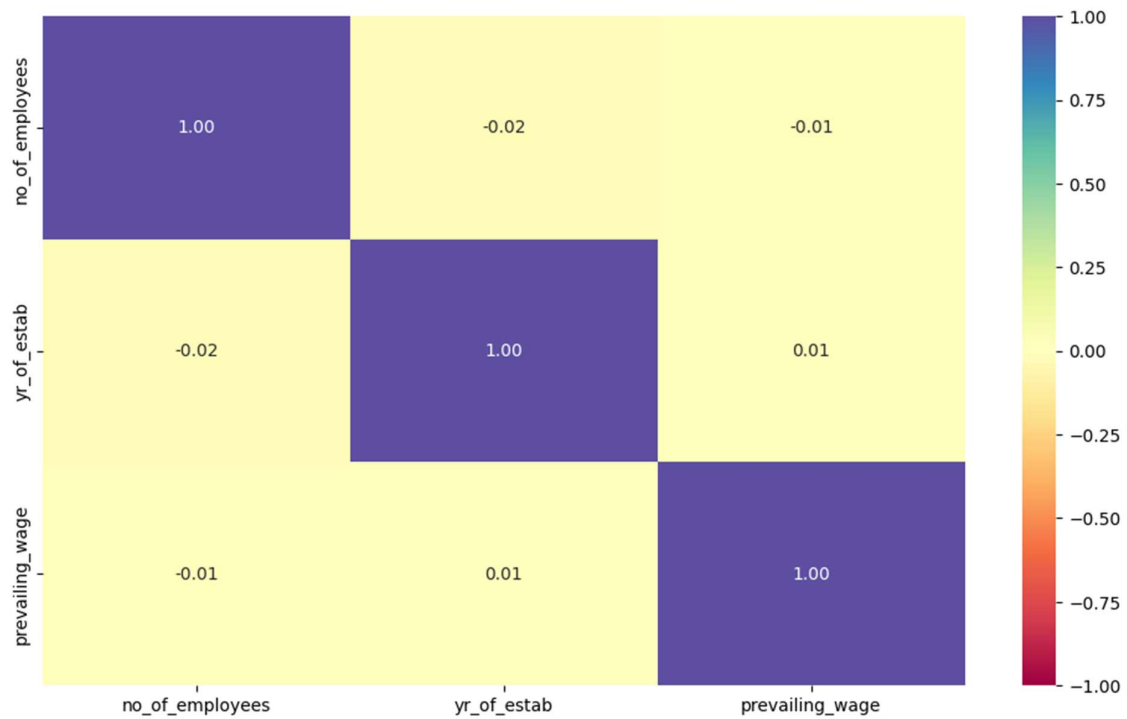


Fig 5.2.2 - Heatmap

- There is no correlation between the numeric variables.

5.2.3 No of employee's vs prevailing wage

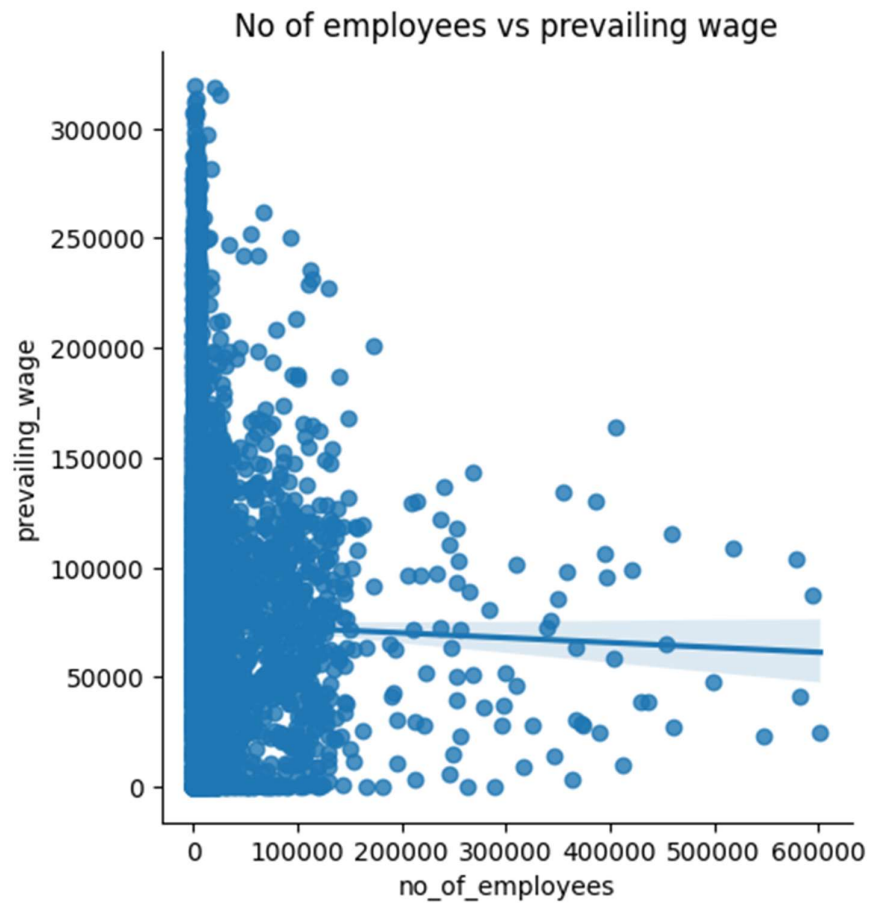


Fig 5.2.3 - No of employees' vs prevailing wage

- There is a slightly negative correlation between no of employees and prevailing wage.
- As the number of employees increases, prevailing wage decreases.

5.2.4 No of employees' vs year of establishment

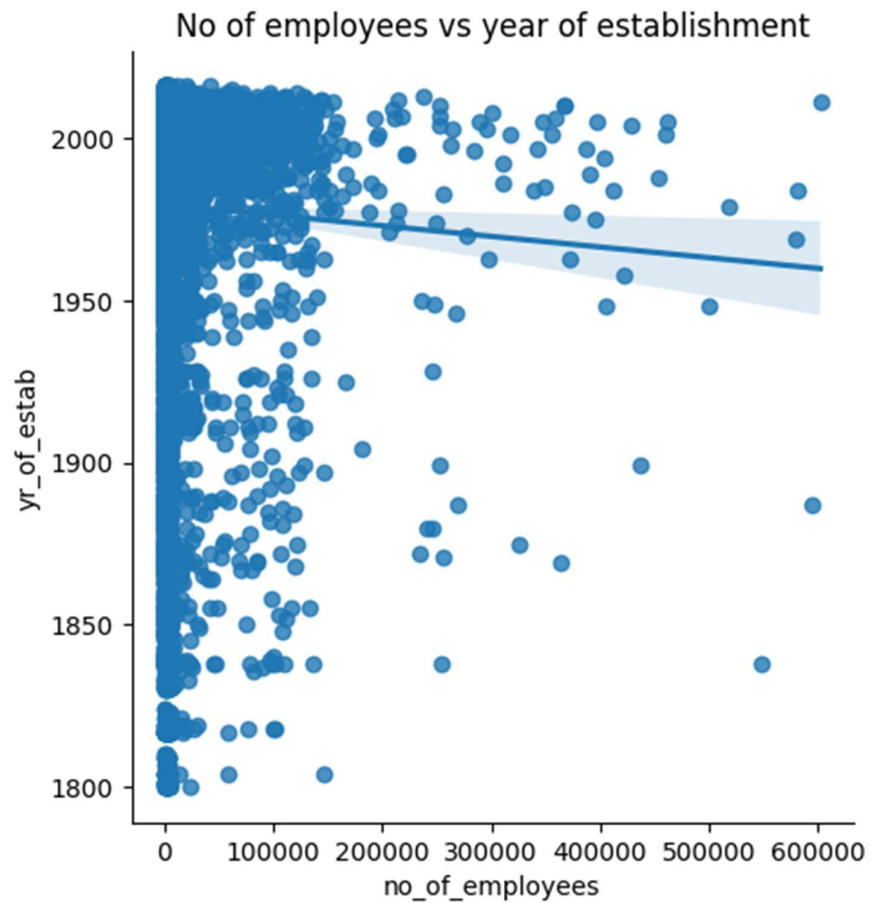


Fig 5.2.4 No of employees' vs year of establishment

- There is a slightly negative correlation between no of employees and year of establishment.
- Most companies are small in size, regardless of when they were established.

5.2.5 Continent vs region of employment

region_of_employment continent	Island	Midwest	Northeast	South	West	All
All	375	4307	7195	7017	6586	25480
Asia	226	3371	4589	4805	3870	16861
North America	82	392	1114	870	834	3292
Europe	32	303	965	973	1459	3732
South America	23	111	266	223	229	852
Africa	8	92	212	102	137	551
Oceania	4	38	49	44	57	192

Table 5.2.5 - Continent vs region of employment

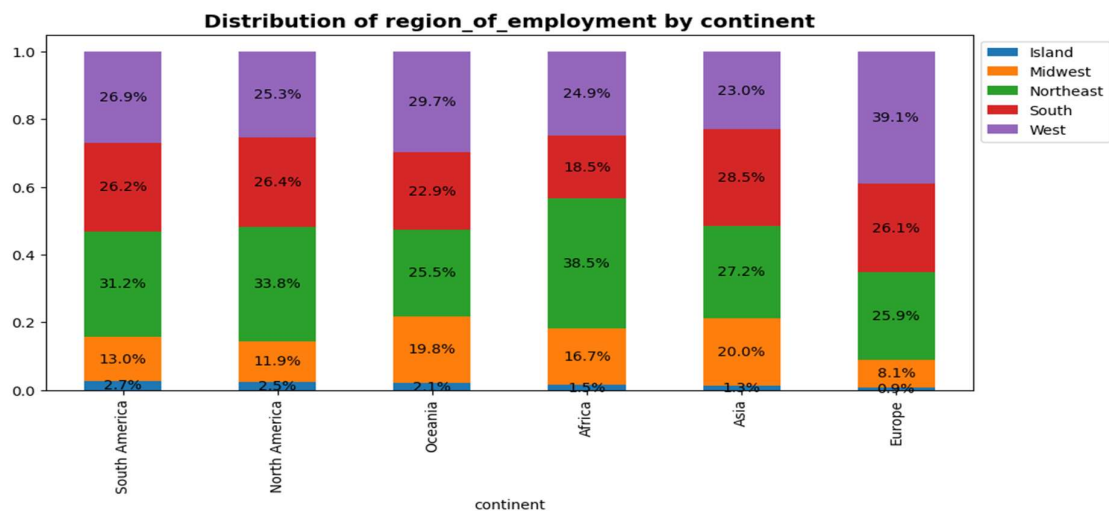


Fig 5.2.5 - Continent vs region of employment

- The West region has a strong presence across all continents and is dominant in Europe.
- The Island region has the least presence across all continent and is dominant in South America.
- Northeast region is dominant in Africa, North America and South America.
- Overall, there is no single uniform region-of-employment pattern across continents.

5.2.6 Has job experience vs requires job training

requires_job_training	N	Y	All
has_job_experience			
All	22525	2955	25480
N	8988	1690	10678
Y	13537	1265	14802

Table 5.2.6 - Has job experience vs requires job training

Distribution of requires_job_training by has_job_experience

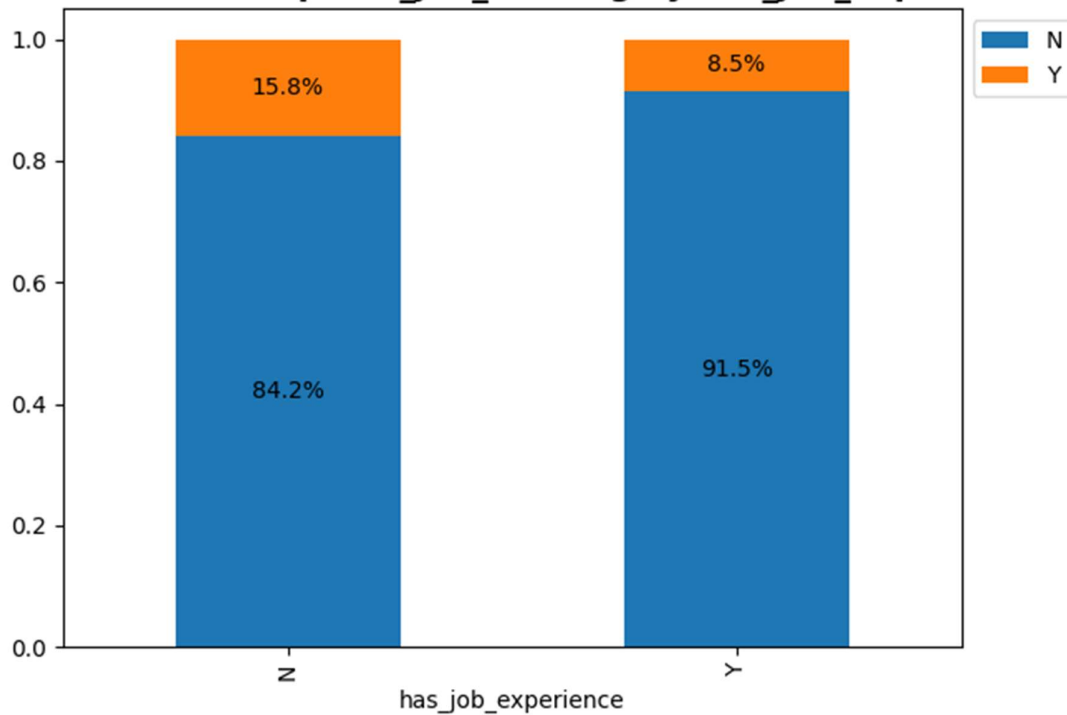


Fig 5.2.6 - Has job experience vs requires job training

- 91.5% of the employees who has job experience do not require job training and 8.5% of the employees require job training.
- 84.2% of the employees who don't have experience do not require job training and 15.8% of the employees require job training.

5.2.7 Education of employee vs full time position

full_time_position	N	Y	All
education_of_employee			
All	2707	22773	25480
Master's	1738	7896	9634
Bachelor's	683	9551	10234
High School	206	3214	3420
Doctorate	80	2112	2192

Table 5.2.7 - Education of employee vs full time position

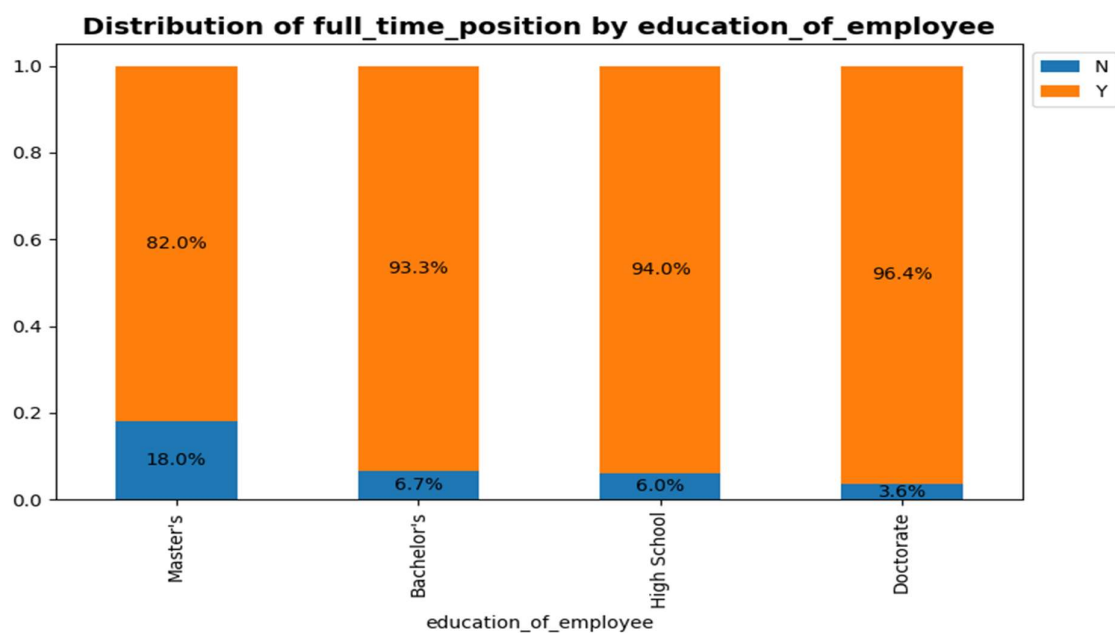


Fig 5.2.7 - Education of employee vs full time position

- 96.4% of Employees who have Doctorate degree have a full-time position, while 3.6% employees have a part time position.
- 94.0% of Employees who have completed High school have a full-time position, while 6.0% employees have a part time position.
- 93.3% of Employees who have Bachelor's degree has a full-time position, while 6.7% employees have a part time position.
- 82.0% of Employees who have Master's degree has a full-time position, while 18.0% employees have a part time position.
- Overall, employees who have higher degree has a full-time position job.

5.2.8 Has job experience vs full time position

full_time_position	N	Y	All
has_job_experience			
All	2707	22773	25480
Y	1410	13392	14802
N	1297	9381	10678

Table 5.2.8 - Has job experience vs full time position

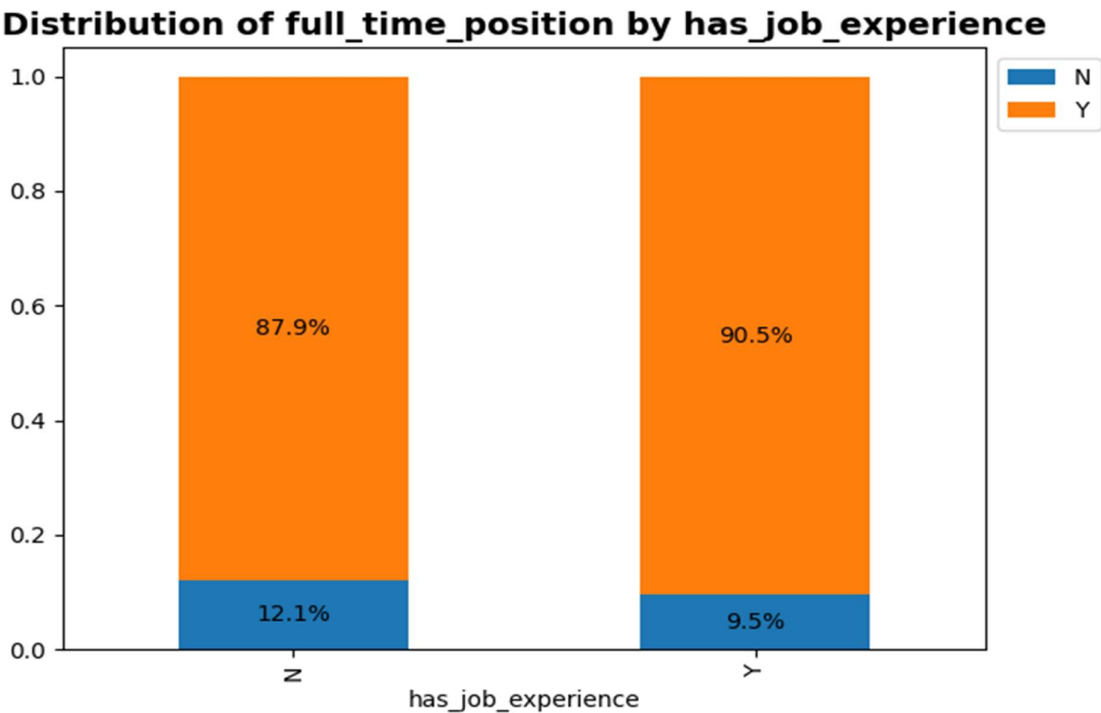


Fig 5.2.8 - Has job experience vs full time position

- Full-time positions are dominant regardless of job experience status.
- Employees with prior job experience have a higher proportion of full-time roles (90.5%) compared to those without experience (87.9%).
- 9.5% of employees with prior experience have a part time position.
- Employees without job experience are slightly more likely to be in part time position(12.1%).

5.2.9 Has job experience vs case status

case_status	Certified	Denied	All
has_job_experience			
All	17018	8462	25480
N	5994	4684	10678
Y	11024	3778	14802

Table 5.2.9 - Has job experience vs case status

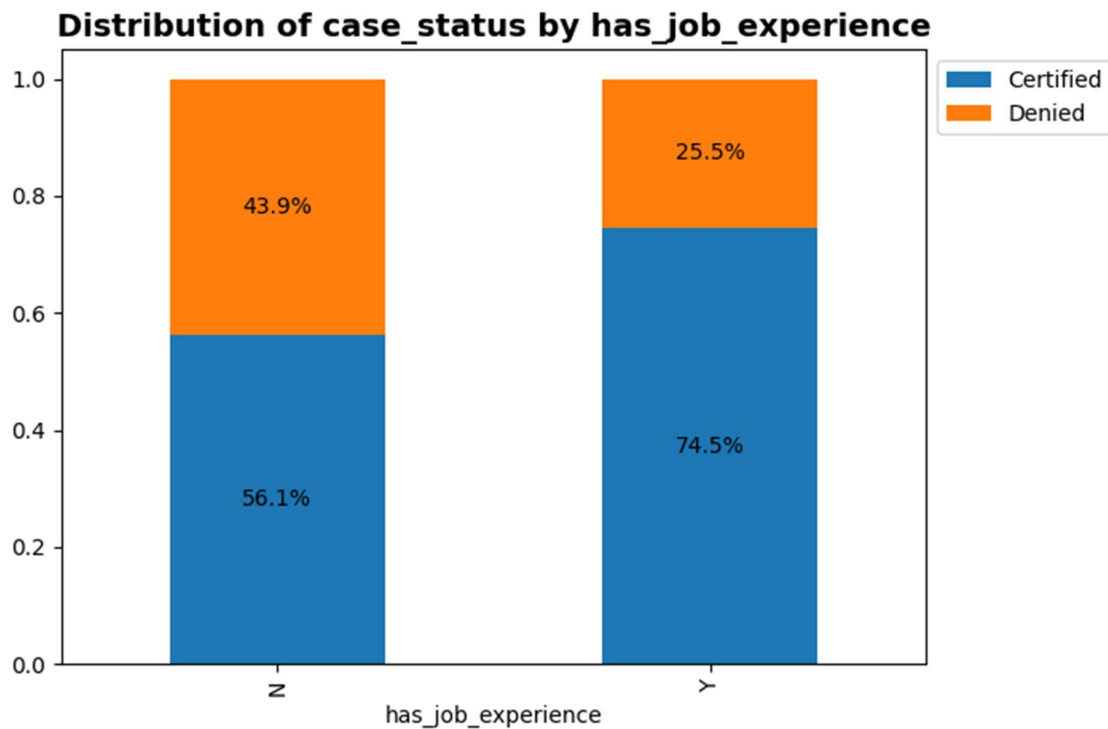


Fig 5.2.9 - Has job experience vs case status

- Among employees who have job experiences, 74.5% employee visa were certified and 25.5% visa were denied.
- Among employees who did not have job experiences, 56.1 % employee visa were certified and 43.9% visa were denied.
- Employees who have job experience have higher chances of getting their visa certified.

5.2.10 Continent vs case status

case_status	Certified	Denied	All
continent			
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192

Table 5.2.10 - Continent vs case status

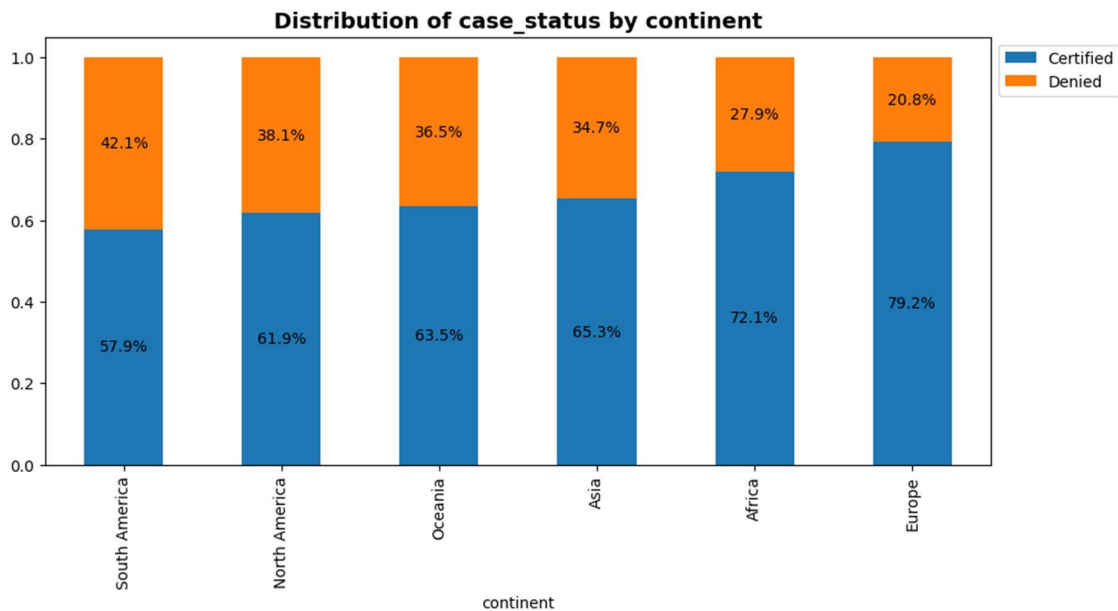


Fig 5.2.10 - Continent vs case status

- Among continents, Europe has the highest proportion(79.2%) of visa getting certified followed by Africa with 72.1%.
- Asia and Oceania fall in the mid-range, with certification rates around 63–65%.
- South America has the most visa denied rates with 42.1%, followed by North America with 38.1%.

5.2.11 Education of employee vs case status

case_status	Certified	Denied	All
education_of_employee			
All	17018	8462	25480
Bachelor's	6367	3867	10234
High School	1164	2256	3420
Master's	7575	2059	9634
Doctorate	1912	280	2192

Table 5.2.11 - Education of employee vs case status

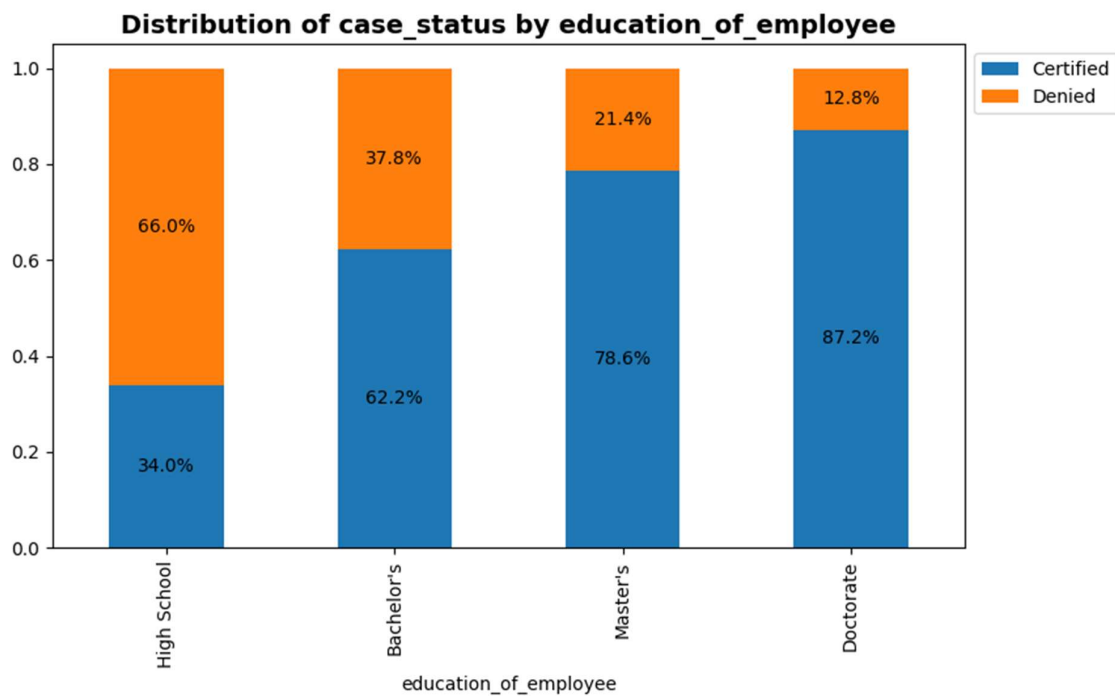


Fig 5.2.11 - Education of employee vs case status

- Employees who have completed Doctorate have higher change of visa getting certified(87.2%) followed by employees who completed Master's(78.6%).
- Employee who has completed high school have a lower visa certified rate(34.0%) and a higher visa denial rate(66.0%).

5.2.12 Full time position vs case status

case_status	Certified	Denied	All
full_time_position			
All	17018	8462	25480
Y	15163	7610	22773
N	1855	852	2707

Table 5.2.12 - Full time position vs case status

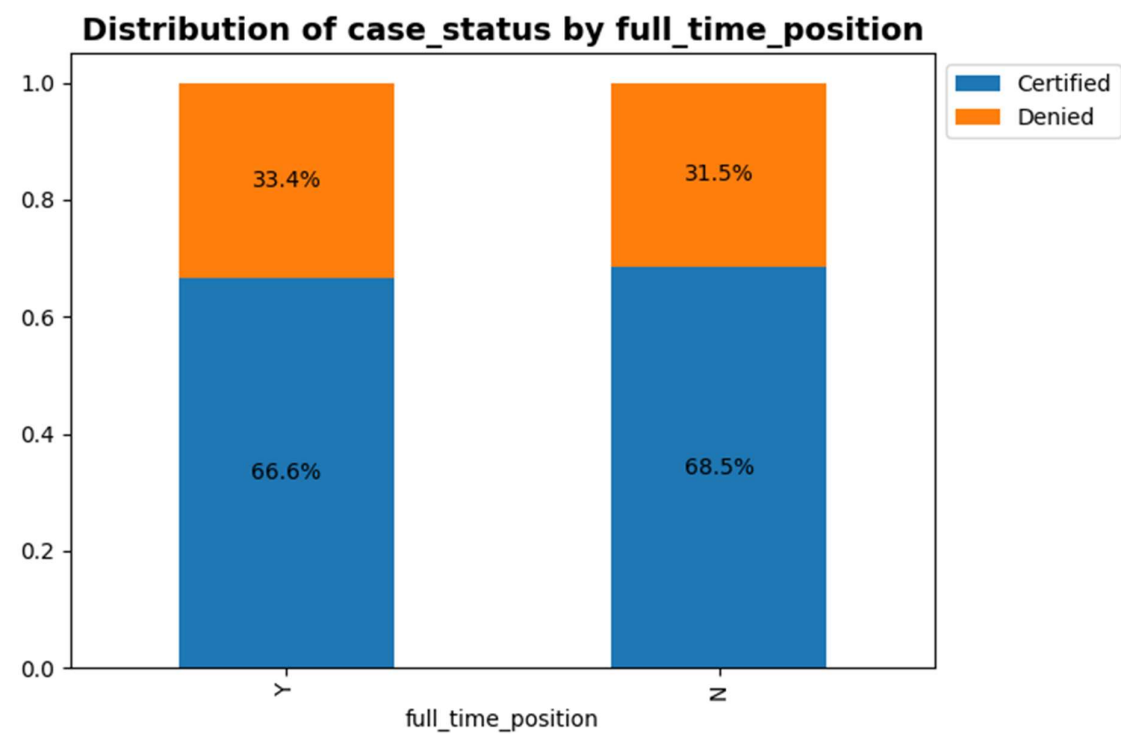


Fig 5.2.12 - Full time position vs case status

- For full time position, 66.7% visas are certified and 33.4% visas are denied.
- For part-time position, 68.5% visas are certified and 31.5% visas are denied.

5.2.13 Region of employment vs case status

case_status	Certified	Denied	All
region_of_employment			
All	17018	8462	25480
Northeast	4526	2669	7195
West	4100	2486	6586
South	4913	2104	7017
Midwest	3253	1054	4307
Island	226	149	375

Table 5.2.13 - Region of employment vs case status

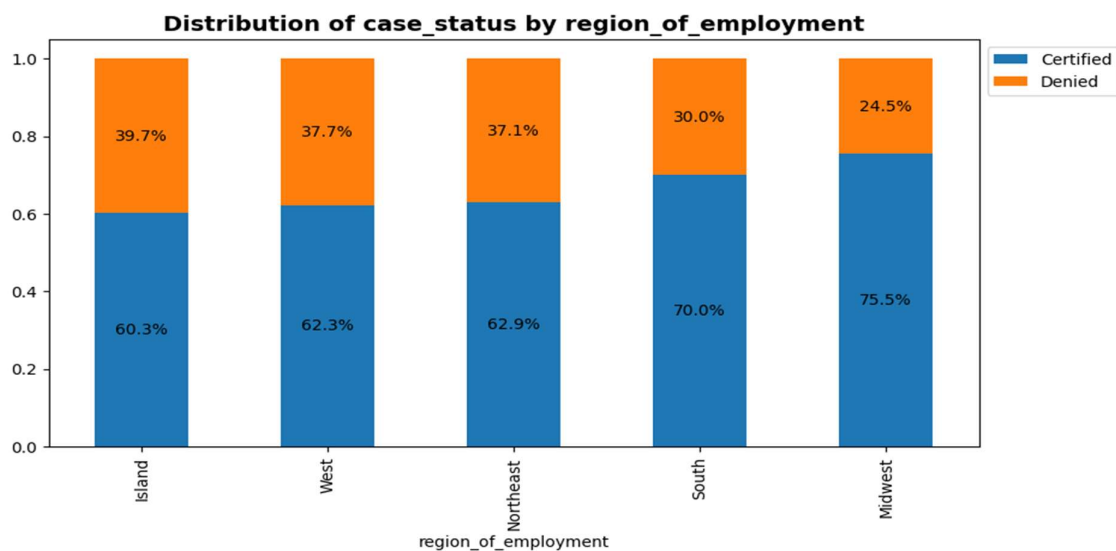


Fig 5.2.13 - Region of employment vs case status

- Midwest region has the highest certification rate(75.5%) and the lowest denial rate(24.5%), followed by South region with certification rate(70.0%) and denial rate(30.0).
- West and Northeast region have almost same certification rate(63%) and denial rate(37%).
- Island region has the highest denial rate(39.7%) when compared to other regions.
- Almost in all the regions, visa certified proportion is higher to denial proportion.

5.2.14 Requires job training vs case status

case_status	Certified	Denied	All
requires_job_training			
All	17018	8462	25480
N	15012	7513	22525
Y	2006	949	2955

Table 5.2.14 - Requires job training vs case status



Fig 5.2.14 - Requires job training vs case status

- Employees who do not require job training has 66.7% visas certified and 33.4% visas denied.
- Employees who require job training has 67.9% visas certified and 32.1% visas denied.
- Visa status is almost same for employees who required job training and who do not require job training.

5.2.15 Distribution of prevailing wage by case status

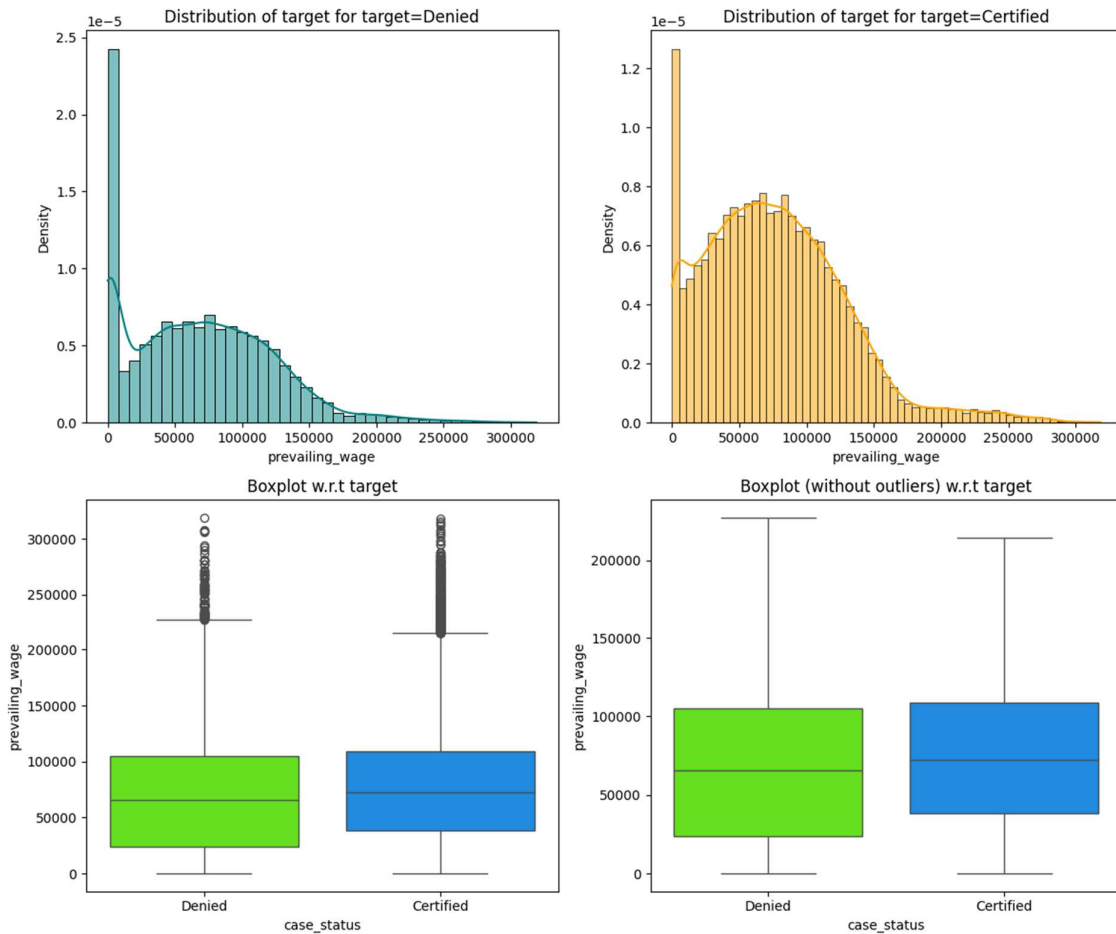


Fig 5.2.15 - Distribution of prevailing wage by case status

- Certified applications tend to have higher prevailing wages than denied applications.
- Both distributions are right-skewed, with long tails toward high wages, which is typical for wage data.
- Even after removing outliers, the wage difference between Certified and Denied remains, confirming the pattern is not driven solely by extreme values.

6. Data Preprocessing

6.1 Duplicate value check

We use duplicated().sum() method to find the duplicated value.

```
np.int64(0)
```

There are no duplicated values in our dataset.

6.2 Missing value treatment

	0
case_id	0
continent	0
education_of_employee	0
has_job_experience	0
requires_job_training	0
no_of_employees	0
yr_of_estab	0
region_of_employment	0
prevailing_wage	0
unit_of_wage	0
full_time_position	0
case_status	0

dtype: int64

Table 6.2 – Missing values

There are no missing values in our dataset. However, in our statistical summary of the dataset, we observed a data error in 'no_of_employees' column.

We'll impute the invalid data after splitting the data into train, validation and test sets to avoid data leakage.

Also, let's go and check for anomalies in other categorical data.

```
continent
Asia      16840
Europe    3727
North America 3287
South America 851
Africa    550
Oceania   192
Name: count, dtype: int64
-----
education_of_employee
Bachelor's 10220
Master's   9622
High School 3416
Doctorate  2189
Name: count, dtype: int64
-----
has_job_experience
Y      14786
N      10661
Name: count, dtype: int64
-----
requires_job_training
N      22498
Y       2949
Name: count, dtype: int64
-----
region_of_employment
Northeast  7189
South      7006
West       6578
Midwest    4299
Island     375
Name: count, dtype: int64
-----
unit_of_wage
Year      22933
Hour      2153
Week       272
Month       89
Name: count, dtype: int64
-----
full_time_position
Y      22741
N       2706
Name: count, dtype: int64
-----
case_status
Certified  17001
Denied     8446
Name: count, dtype: int64
-----
```

From the above information, we can observe that there are no anomalies present in other variables.

6.3 Outlier check

Let's check outliers for all the numerical variables.

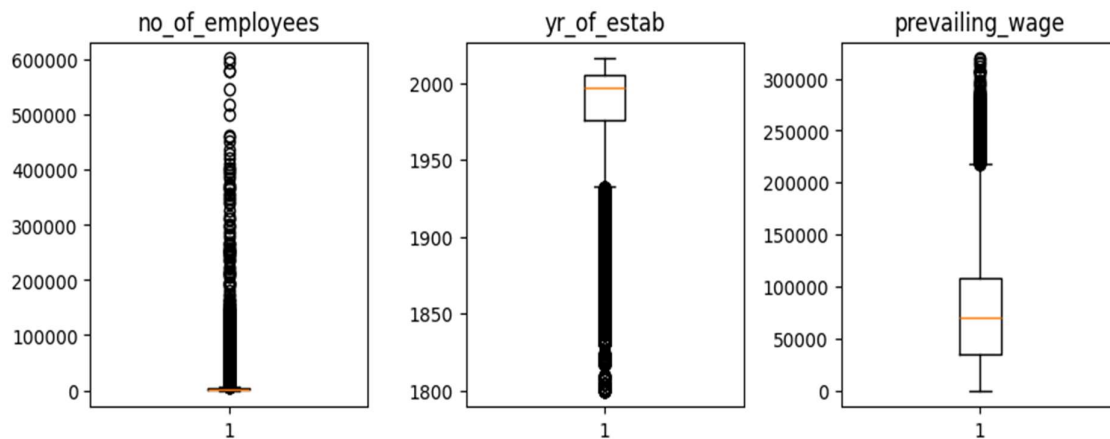


Fig 6.3 – Outlier check

- There are few outliers. However, we do not treat them as they are genuine data points.

6.4 Feature Engineering

1. Since 'case_id' variable consists only of unique ID's, we'll drop this variable.
2. We'll encode the target variable. Replace Certified as 0 and Denied as 1.

6.5 Data preparation for modelling

- We want to facilitate the visa process and recommend a suitable profile for the applicants for whom the visa should be certified or denied.
- Before we proceed to build a model, we'll have to encode categorical features.
- We'll split the data into training, validation and test to be able to evaluate the model that we build on the train data.

Before we prepare data for modelling, let's check the value counts of our target variable.

proportion	
case_status	
1	0.668
0	0.332
dtype: float64	

Table 6.5.1 – Value counts of case status

- 67% of the visas were certified
- 33% of the visas were denied
- Clearly, our dataset is imbalanced.

In our dataset, we have three categorical (binary, ordinal and nominal) data's. Before splitting the data, we'll encode binary variable using binary encoding, ordinal variable using label encoding.

'has_job_experience', 'requires_job_training', 'full_time_position' are binary variables. We'll replace Y as 1 and N as 0.

'education_of_employee' is an ordinal variable. There we'll do a label encoding.

High School: 0
Bachelor's: 1
Master's: 2
Doctorate: 3

We'll define our X and y variable. X consists of all independent variables and y consists of our target variable.

We'll split our data into training, validation and test sets.

```
Number of rows in train data = (12740, 10)
Number of rows in validation data = (7644, 10)
Number of rows in test data = (5096, 10)
```

After splitting the data, we'll treat the invalid data in the 'no_of_employees' column. We'll replace the negative values with np.nan and use SimpleImputer preprocessing tool to impute the nan values with median in train, validation and test sets.

```
Min value of employees in train: 12.0
Min value of employees in validation: 12.0
Min value of employees in test: 16.0
```

Then we proceed to do one-hot encoding on the variables, 'continent', 'region_of_employment', 'unit_of_wage' using pd.get_dummies.

```
Number of rows and columns in train data after encoding = (12740, 19)
Number of rows and columns in validation data after encoding = (7644, 19)
Number of rows and columns in test data after encoding = (5096, 19)
```

7. Model building – Original data

7.1 Model evaluation criterion

Model can make wrong predictions as:

1. Predicting the visa application will be certified, but in reality, the application is denied.
2. Predicting the visa application will be denied, but in reality, the application is certified.

Which case is more important?

- If a visa is certified when it has to be denied, the wrong employee will get the job position while US citizens will miss the opportunity to work in that position.
- If a visa is denied when it has to be certified the US will lose a suitable employee that can contribute to the economy.
- Both the cases are important.

How to reduce this loss

- OFLC would want F1 score to be maximized, the greater the F1 score, the higher the chances of minimizing false positives and false negatives.
- Secondary evaluation metrics will include **Accuracy, Precision, Recall**, and the **Confusion Matrix** to provide a comprehensive understanding of the model's performance:
 - **Accuracy:** Overall correctness of the model.
 - **Precision:** The proportion of applicants predicted to be certified (approved) who were actually certified. (minimizing false positives).
 - **Recall:** The proportion of applicants who were actually certified (approved) that the model correctly predicted as certified. (minimizing false negatives).
 - **Confusion Matrix:** A detailed breakdown of true positives, true negatives, false positives, and false negatives.
- We created functions to calculate different metrics and confusion matrix

7.2 Initial Model Building

We create five models with our original dataset. We build Decision tree, Bagging, Random Forest, AdaBoost and Gradient Boosting models. Since our dataset is imbalanced, we use `class_weight='balanced'` parameter while building Decision tree, Bagging and Random Forest.

We train the model with our training set and evaluate it with validation set.

--- All Model Performance on Training Data ---

	Accuracy	Recall	Precision	F1
dtree	1.000	1.000	1.000	1.000
Bagging	0.983	0.985	0.991	0.988
Random forest	1.000	1.000	1.000	1.000
GBM	0.756	0.879	0.782	0.828
Adaboost	0.737	0.883	0.761	0.818

Table 7.2.1 – Original data – Training set

--- All Model Performance on Validation Data ---

	Accuracy	Recall	Precision	F1
dtree	0.669	0.754	0.752	0.753
Bagging	0.708	0.794	0.774	0.784
Random forest	0.725	0.850	0.765	0.805
GBM	0.755	0.883	0.780	0.828
Adaboost	0.737	0.889	0.759	0.819

Table 7.2.2 – Original data – Validation set

Since our goal is to maximize f1-score, we will find the difference of f1-score between training and validation set.

Training and Validation Performance Difference:

dtree: Training Score: 1.000, Validation Score: 0.753, Difference: 0.247
Bagging: Training Score: 0.988, Validation Score: 0.784, Difference: 0.204
Random forest: Training Score: 1.000, Validation Score: 0.805, Difference: 0.195
GBM: Training Score: 0.828, Validation Score: 0.828, Difference: -0.000
Adaboost: Training Score: 0.818, Validation Score: 0.819, Difference: -0.001

Observations:

1. Decision Tree(dtrees)

- The model performs perfectly on training data but slightly lower in validation data which indicates of overfitting. The decision tree has memorized the training data but fails to
- generalize well.

2. Bagging(BaggingClassifier with Decision tree):

- Training score is high and the validation score is better than the decision tree. But the difference indicates of overfitting.

3. Random Forest(RandomForestClassifier)

- The model performs extremely well on train data but slightly poor on validation data indicating overfitting.

4. Gradient Boosting:

- The model training score is good, but not perfect. The validation score is very close to the training score which indicates that the model generalizes well and not overfitting.

5. AdaBoost:

Similar to GBM, AdaBoost model also has a non-perfect training score. But it is comparable to validation score. This model also generalizes well and does not overfit.

GBM and AdaBoost are clearly the best-performing models in terms of generalization and avoiding overfitting.

8. Model Building - Oversampled Data

In oversampling, we increase the minority samples(Denied) using SMOTE technique.

```
Before Oversampling, counts of label 'Certified': 8509
Before Oversampling, counts of label 'Denied': 4231
```

```
After Oversampling, counts of Certified 'Certified': 8509
After Oversampling, counts of label 'Denied': 8509
```

```
After Oversampling, the shape of train_X: (17018, 19)
After Oversampling, the shape of train_y: (17018,)
```

We build five models – Decision Tree, Bagging, Random Forest, AdaBoost and Gradient Boosting with the oversampled data.

--- All Model Performance on Oversampled Training Data ---

	Accuracy	Recall	Precision	F1	
dtree	1.000	1.000	1.000	1.000	
Bagging	0.987	0.982	0.992	0.987	
Random forest	1.000	1.000	1.000	1.000	
GBM	0.812	0.866	0.782	0.822	
Adaboost	0.792	0.865	0.755	0.806	

Table 8.1 – Oversampled – Training set

--- All Model Performance on Oversampled Validation Data ---

	Accuracy	Recall	Precision	F1
dtree	0.655	0.734	0.746	0.740
Bagging	0.707	0.782	0.780	0.781
Random forest	0.721	0.834	0.768	0.800
GBM	0.752	0.872	0.782	0.825
Adaboost	0.738	0.877	0.765	0.818

Table 8.2 – Oversampled – Validation set

The difference of f1-score between training and validation set of oversampled data is:

Training and Validation Performance Difference for oversampled data:

dtree: Training Score: 1.000, Validation Score: 0.740, Difference: 0.260
Bagging: Training Score: 0.987, Validation Score: 0.781, Difference: 0.206
Random forest: Training Score: 1.000, Validation Score: 0.800, Difference: 0.200
GBM: Training Score: 0.822, Validation Score: 0.825, Difference: -0.003
Adaboost: Training Score: 0.806, Validation Score: 0.818, Difference: -0.011

Observations:

1. Decision Tree(dtrees)

- The model performs perfectly on training data but slightly lower in validation data which indicates of overfitting. The decision tree has memorized the training data but fails to generalize well.

2. Bagging(BaggingClassifier with Decision tree):

- Training score is high and the validation score is lower than the decision tree. But the difference indicates of overfitting.

3. Random Forest(RandomForestClassifier)

- The model performs well on train data but slightly poor on validation data indicating overfitting.

4. Gradient Boosting:

- The model training score is good, but not perfect. The validation score is very close to the training score which indicates that the model generalizes well and not overfitting.

5. AdaBoost:

Similar to GBM, AdaBoost model also has a non-perfect training score. But it is comparable to validation score. This model also generalizes well and does not overfit.

Oversampling did not significantly help with Decision Tree, Bagging, Random forest with their generalization issues.

Though GBM and AdaBoost training score and validation score dropped slightly from the initial model, their performance remained stable and consistent and are clearly the best-performing models.

9. Model Building – Undersampled Data

In undersampling, we remove random records from the majority class using RandomUnderSampler technique.

```
Before Under Sampling, counts of label 'Certified': 8509
Before Under Sampling, counts of label 'Denied': 4231
```

```
After Under Sampling, counts of label 'Certified': 4231
After Under Sampling, counts of label 'Denied': 4231
```

```
After Under Sampling, the shape of train_X: (8462, 19)
After Under Sampling, the shape of train_y: (8462,)
```

We again build five models – Decision Tree, Bagging, Random Forest, AdaBoost and Gradient Boosting with undersampled data. We'll train the models with training set and evaluate it with validation set.

--- All Model Performance on Undersampled Training Data ---

	Accuracy	Recall	Precision	F1
dtree	1.000	1.000	1.000	1.000
Bagging	0.979	0.967	0.990	0.979
Random forest	1.000	1.000	1.000	1.000
GBM	0.723	0.710	0.729	0.720
Adaboost	0.691	0.705	0.685	0.695

Table 9.1 – Undersampled – Training set

--- All Model Performance on Undersampled Validation Data ---

	Accuracy	Recall	Precision	F1
dtree	0.634	0.634	0.778	0.698
Bagging	0.651	0.622	0.811	0.704
Random forest	0.680	0.675	0.814	0.738
GBM	0.704	0.702	0.828	0.760
Adaboost	0.700	0.716	0.812	0.761

Table 9.2 – Undersampled – Validation set

The difference of f1-score between training and validation set of undersampled data is:

Training and Validation Performance Difference of undersampled data:

```
dtree: Training Score: 1.000, Validation Score: 0.698, Difference: 0.302
Bagging: Training Score: 0.979, Validation Score: 0.704, Difference: 0.275
Random forest: Training Score: 1.000, Validation Score: 0.738, Difference: 0.262
GBM: Training Score: 0.720, Validation Score: 0.760, Difference: -0.040
Adaboost: Training Score: 0.695, Validation Score: 0.761, Difference: -0.066
```

Observations:

1. Decision Tree(dtrees)

- The model performs perfectly on training data but slightly lower in validation data which indicates of overfitting. The decision tree has memorized the training data but fails to generalize well.

2. Bagging(BaggingClassifier with Decision tree):

- Training score is high and the validation score is better than the decision tree. But the difference indicates of overfitting.

3. Random Forest(RandomForestClassifier)

- The model performs well on train data but slightly poor on validation data indicating overfitting.

4. Gradient Boosting:

- The model training score(0.72) is significantly lower than (original:0.828, oversampled:0.822). Also, the validation score(0.76) dropped(original:0.828, oversampled:0.825). While the training and validation scores are close indicating generalization, the overall F1 scores have decreased.

5. AdaBoost:

- The model training score(0.695) is significantly lower than (original:0.818, oversampled:0.806). Also, the validation score(0.761) dropped(original:0.819, oversampled:0.818). Similar to GBM, the training and validation scores are close indicating generalization, the overall F1 scores have decreased.

For all models, undersampling has led to a noticeable drop in F1 scores on both the training and validation sets. Undersampling appears to be less effective for this dataset compared to using the original imbalanced data or oversampled data.

10. Model Performance Improvement using Hyperparameter Tuning

- After building 15 models, it was observed that both AdaBoost and GBM models, trained on original and oversampled dataset, exhibited strong performance on both the training and validation datasets.
- Sometimes models might overfit after undersampling and oversampling, so it's better to tune the models to get a generalized performance.
- We will tune these 4 models using the same data (original or oversampled) as we trained them on before.
- We'll use '**RandomizedSearchCV**', a hyperparameter tuning technique in scikit-learn that allows you to search over a specified parameter distribution.

10.1 Tuning AdaBoostClassifier model with original data

We use '**best_params_**' which returns a dictionary containing the best set of parameters.

- ✓ estimator=DecisionTreeClassifier(max_depth=3, min_samples_leaf=10, random_state=1)
- ✓ n_estimators=20
- ✓ learning_rate=0.2
- ✓ random_state=1

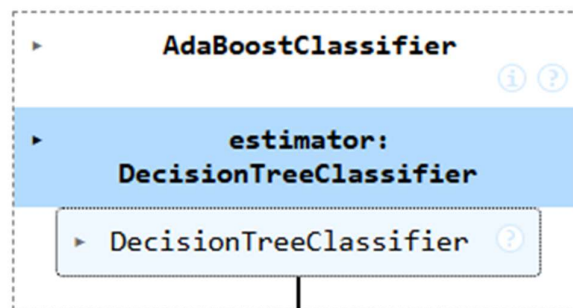


Fig 10.1 – AdaBoost model tuned on original data

10.1a Checking model performance on training set

	Accuracy	Recall	Precision	F1
0	0.740	0.921	0.748	0.825

Table 10.1a - Training performance of AdaBoost tuned on original data

10.1b Checking model performance on validation set

	Accuracy	Recall	Precision	F1
0	0.738	0.923	0.746	0.825

Table 10.1b - Validation performance of AdaBoost tuned on original data

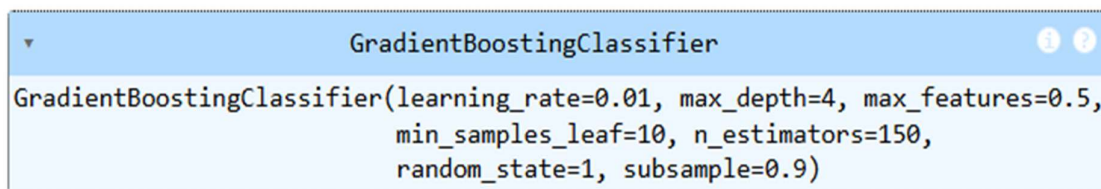
Observations:

- After tuning hyperparameter, the f1 score of training and validation has increased slightly from the original model.
- Tuned AdaBoost model is performing well and is able to generalize well.

10.2 Tuning GradientBoostingClassifier model with original data

We tune the Gradient Boosting model (original data) with the following best parameters found by RandomizedSearchCV.

- ✓ random_state=1,
- ✓ subsample= 0.9,
- ✓ n_estimators=150,
- ✓ min_samples_leaf=10
- ✓ max_features=0.5,
- ✓ max_depth=4,
- ✓ learning_rate=0.01



```
GradientBoostingClassifier(learning_rate=0.01, max_depth=4, max_features=0.5,
                           min_samples_leaf=10, n_estimators=150,
                           random_state=1, subsample=0.9)
```

Fig 10.2 Gradient boosting tuned on original data

10.2a Checking model performance on training set

	Accuracy	Recall	Precision	F1
0	0.740	0.920	0.749	0.825

Table 10.2a - Training performance of GBM tuned on original data

10.2b Checking model performance on validation set

	Accuracy	Recall	Precision	F1
0	0.735	0.920	0.744	0.823

Table 10.2ba - Validation performance of GBM tuned on original data

Observations:

- After tuning, Gradient Boosting is performing and generalizing well, but still have low f1 scores than Gradient boosting trained on original data.
- Tuning didn't improve performance but the model is stable.

10.3 Tuning AdaBoostClassifier model with oversampled data

The best parameters for the model are:

- ✓ estimator=DecisionTreeClassifier(max_depth=3, min_samples_leaf=5, random_state=1)
- ✓ n_estimators=30
- ✓ learning_rate=1
- ✓ random_state=1

We tune the model using the above parameters.

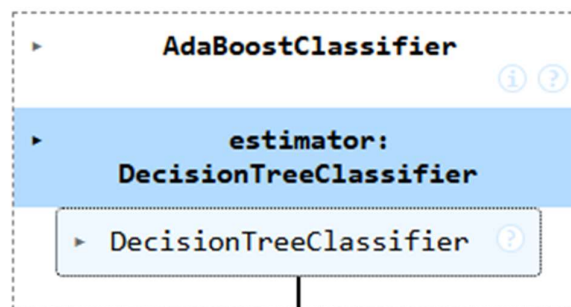


Fig 10.3 AdaBoost tuned on oversampled data

10.3a Checking model performance on Training set

	Accuracy	Recall	Precision	F1
0	0.804	0.851	0.778	0.813

Table 10.3a - Training performance of AdaBoost tuned on oversampled data

10.3b Checking model performance on validation set

	Accuracy	Recall	Precision	F1
0	0.742	0.857	0.779	0.816

Table 10.3b - Validation performance of AdaBoost tuned on oversampled data

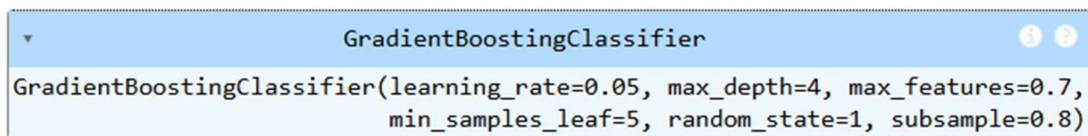
Observations:

- AdaBoost tuned trained with oversampled data is performing slightly lesser than AdaBoost model trained with original data.
- However, AdaBoost model tuned on oversampled data is performing well and is able to generalize well.

10.4 Tuning GradientBoostingClassifier model with oversampled data

We tune the Gradient Boosting model (oversampled data) with the following best parameters found by RandomizedSearchCV.

- ✓ random_state=1,
- ✓ subsample= 0.8,
- ✓ n_estimators=100,
- ✓ min_samples_leaf=5,
- ✓ max_features=0.7,
- ✓ max_depth=4,
- ✓ learning_rate=0.05



```
GradientBoostingClassifier(learning_rate=0.05, max_depth=4, max_features=0.7,
                           min_samples_leaf=5, random_state=1, subsample=0.8)
```

Fig 10.4 GBM tuned on oversampled data

10.4a Checking model performance on Training set

	Accuracy	Recall	Precision	F1
0	0.812	0.868	0.781	0.822

Table 10.4a - Training performance of GBM tuned on oversampled data

10.4b Checking model performance on Validation set

	Accuracy	Recall	Precision	F1
0	0.755	0.874	0.784	0.827

Table 10.4b - Validation performance of GBM tuned on oversampled data

11. Model Performance Comparison and Final Model Selection

11.1 Training Performance Comparison

Training performance comparison:

	Accuracy	Recall	Precision	F1
AdaBoost (Original Data) - Tuned	0.740	0.921	0.748	0.825
GBM (Original Data) - Untuned	0.756	0.879	0.782	0.828
GBM (Original Data) - Tuned	0.740	0.920	0.749	0.825
AdaBoost (Oversampled Data) - Tuned	0.804	0.851	0.778	0.813
GBM (Oversampled Data) - Tuned	0.812	0.868	0.781	0.822

Table 11.1 – Training performance for all models

- In the training, the Gradient Boosting model trained on original data is performing well.
- It is followed by AdaBoost model trained on original data and Gradient Boosting model tuned on original data.

11.2 Validation Performance Comparison

Validation performance comparison:

	Accuracy	Recall	Precision	F1
AdaBoost (Original Data) - Tuned Validation	0.738	0.923	0.746	0.825
GBM (Original Data) - Untuned Validation	0.755	0.883	0.780	0.828
GBM (Original Data) - Tuned Validation	0.735	0.920	0.744	0.823
AdaBoost (Oversampled Data) - Tuned Validation	0.742	0.857	0.779	0.816
GBM (Oversampled Data) - Tuned Validation	0.755	0.874	0.784	0.827

Table 11.2 – Validation performance for all models

- In validation, Gradient Boosting model with original data is performing well, followed by Gradient Boosting model tuned with oversample data, AdaBoost model tuned with original data.
- All the models are able to generalize well.

11.3 Test Performance Comparison

Let's check the test results for the five models.

```
Original GBM (Untuned) Test:
Accuracy Recall Precision F1
0 0.706 0.703 0.831 0.762
```

Table 11.3a – Original GBM(untuned) Test

```
AdaBoost (Oversampled Data) - Tuned Test:
Accuracy Recall Precision F1
0 0.737 0.853 0.775 0.812
```

Table 11.3b – AdaBoost tuned (oversampled data) Test

```
GBM (Oversampled Data) - Tuned Test:
Accuracy Recall Precision F1
0 0.743 0.866 0.776 0.818
```

Table 11.3c – GBM tuned (oversampled) Test

```
GBM (Original Data) - Tuned Test:
Accuracy Recall Precision F1
0 0.731 0.919 0.740 0.820
```

Table 11.3d – GBM tuned(Original) Test

```
AdaBoost (Original Data) - Tuned Test:
Accuracy Recall Precision F1
0 0.733 0.921 0.741 0.822
```

Table 11.3e– AdaBoost tuned (original) Test

Test performance comparison:

	Accuracy	Recall	Precision	F1
Original GBM (Untuned) Test	0.706	0.703	0.831	0.762
AdaBoost (Original Data) - Tuned Test	0.733	0.921	0.741	0.822
GBM (Original Data) - Tuned Test	0.731	0.919	0.740	0.820
GBM (Oversampled Data) - Tuned Test	0.743	0.866	0.776	0.818
AdaBoost (Oversampled Data) - Tuned Test	0.737	0.853	0.775	0.812

Table 11.3f – Testing performance for all models

- Though Grading Boosting model with original data was performing well in training and validation, the f1 score has dropped to 0.762 for unseen data. Also, there is an increase in Precision and a decrease in Recall.
- It seems that AdaBoost model tuned on original data is performing well on unseen data with highest f1-score and recall score.
- The best model is AdaBoost model tuned on original data with 82.2% on the test set followed by Gradient Boosting model tuned on original data with high f1 score and recall score.

11.4 Confusion matrix

Let us plot confusion matrix for our tuned AdaBoost model (original data) and see how it looks in training, validation and test sets.

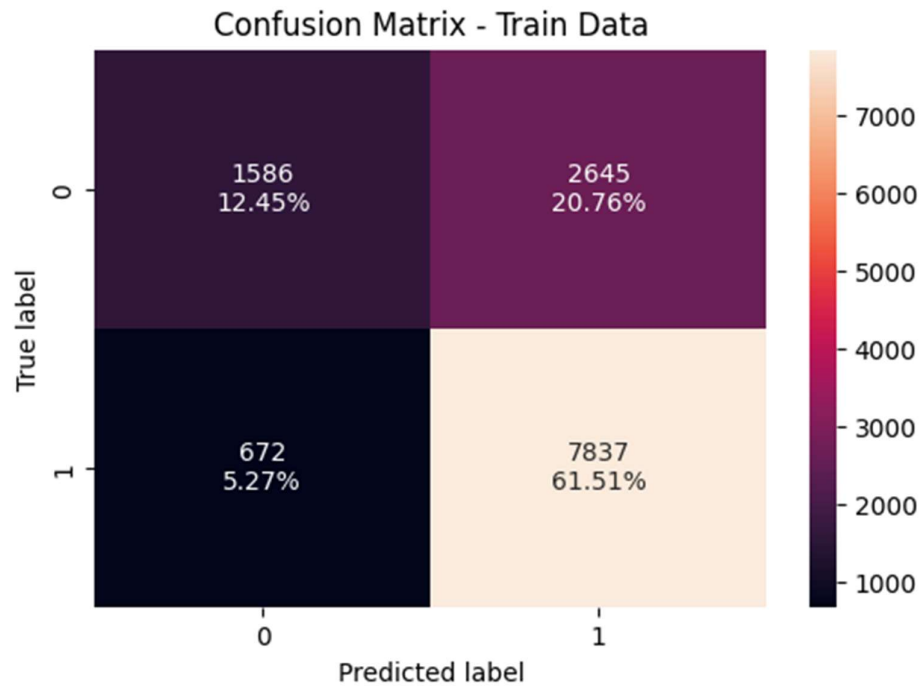


Fig 11.4a – Confusion Matrix for AdaBoost(Tuned) – Train data

- True Negatives (TN): 1586 (Correctly predicted Denied)
- False Positives (FP): 2645 (Incorrectly predicted as Certified, but actually Denied)
- False Negatives (FN): 672 (Incorrectly predicted as Denied, but actually Certified)
- True Positives (TP): 7837 (Correctly predicted as Certified)

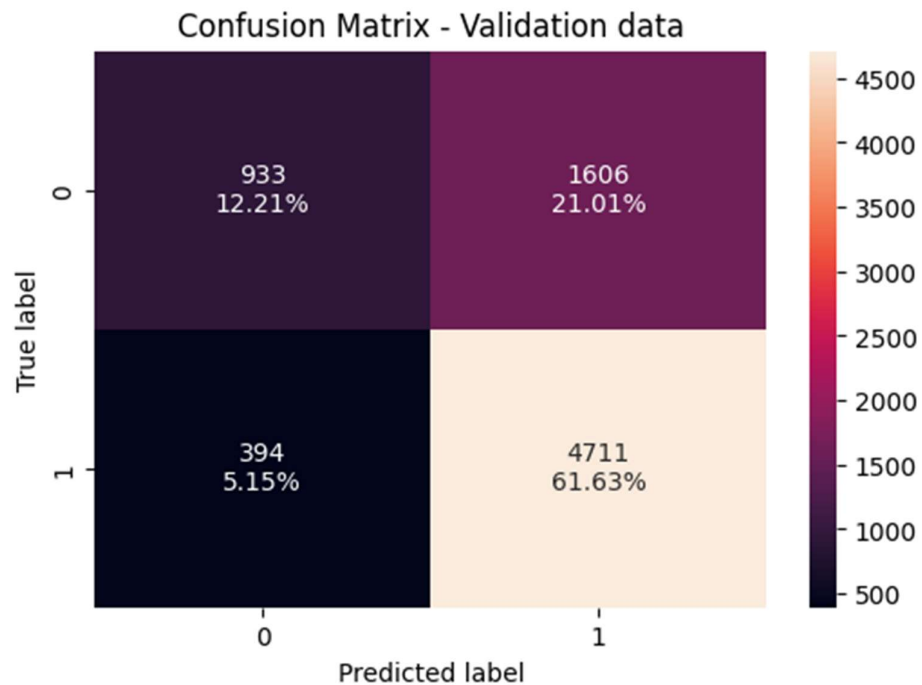


Fig 11.4b – Confusion Matrix for AdaBoost(Tuned) – Validation data

- True Negatives (TN): 933 (Correctly predicted Denied)
- False Positives (FP): 1606 (Incorrectly predicted as Certified, but actually Denied)
- False Negatives (FN): 394 (Incorrectly predicted as Denied, but actually Certified)
- True Positives (TP): 4711 (Correctly predicted as Certified)

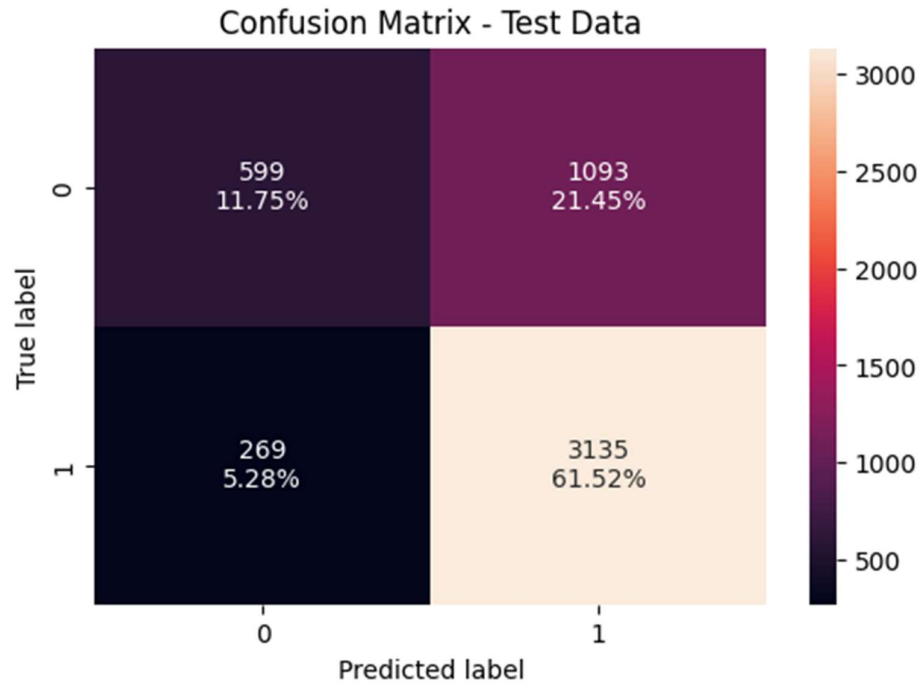


Fig 11.4c – Confusion Matrix for AdaBoost(Tuned) – Test data

- True Negatives (TN): 599 (Correctly predicted Denied)
- False Positives (FP): 1093 (Incorrectly predicted as Certified, but actually Denied)
- False Negatives (FN): 269 (Incorrectly predicted as Denied, but actually Certified)
- True Positives (TP): 3135 (Correctly predicted as Certified)

11.5 Precision-Recall curve

We import `precision_recall_curve` from `sklearn.metrics` library. The PR curve allows comparison of model performance across all decision thresholds rather than at a single cutoff.

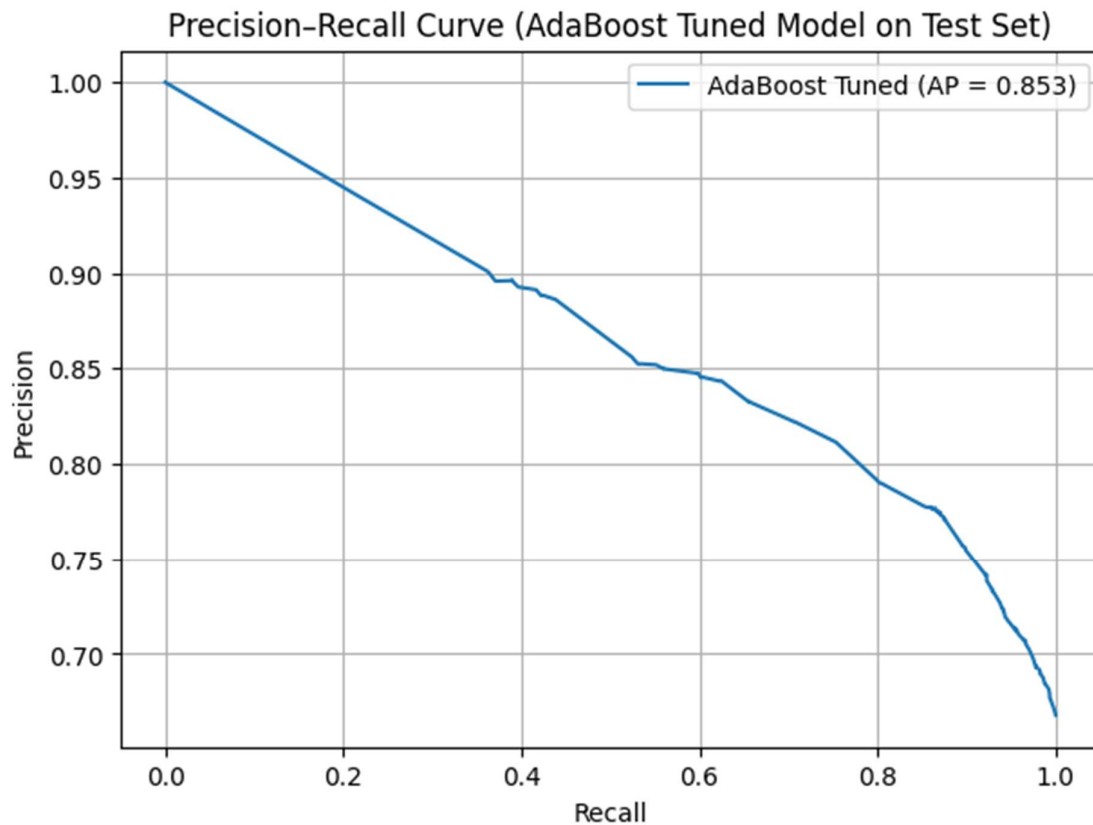


Fig 11.5a – Precision-Recall curve

We'll plot Precision–Recall curve to compare AdaBoost model tuned on original data and Gradient Boosting model tuned on original data because the dataset is imbalanced and precision and recall are more informative than accuracy.

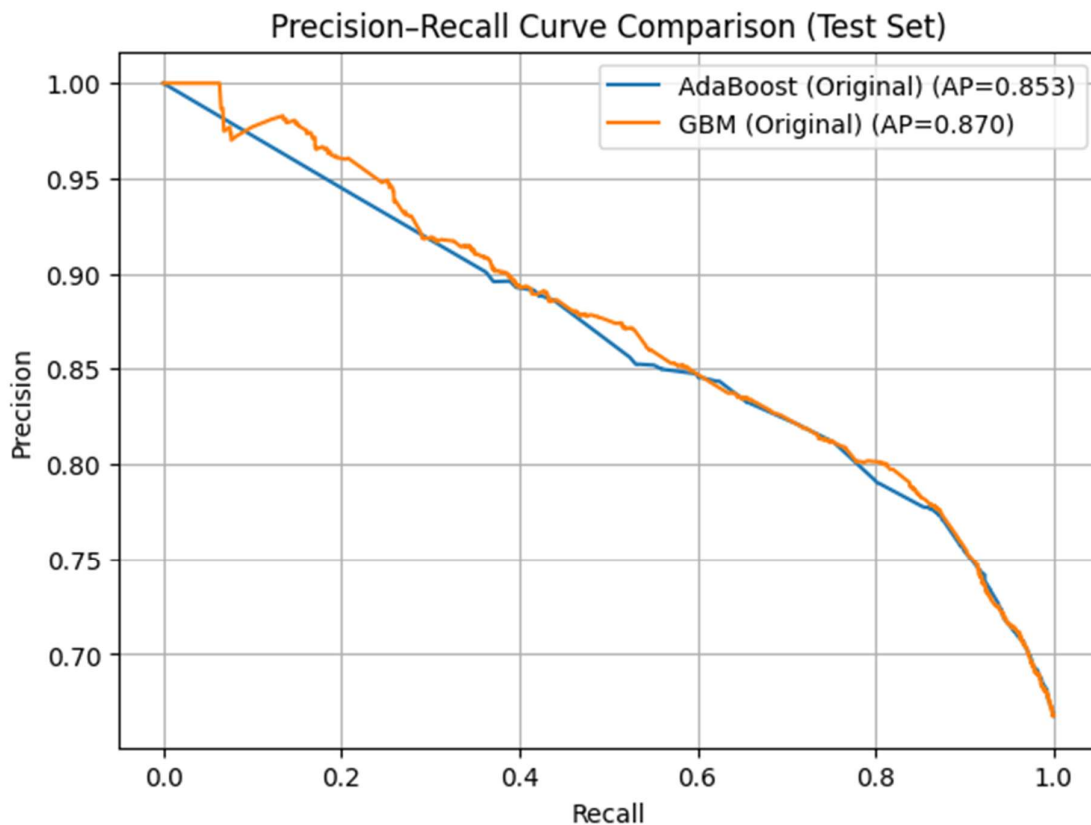


Fig 11.5b Precision-Recall curve comparison(Test set)

- While the tuned AdaBoost model achieves the highest F1-score on the test set at the selected classification threshold, the tuned Gradient Boosting model attains a higher average precision, indicating stronger overall precision–recall trade-offs across thresholds.
- AdaBoost is best at the chosen threshold; GBM is better across thresholds.

11.6 Feature Importance

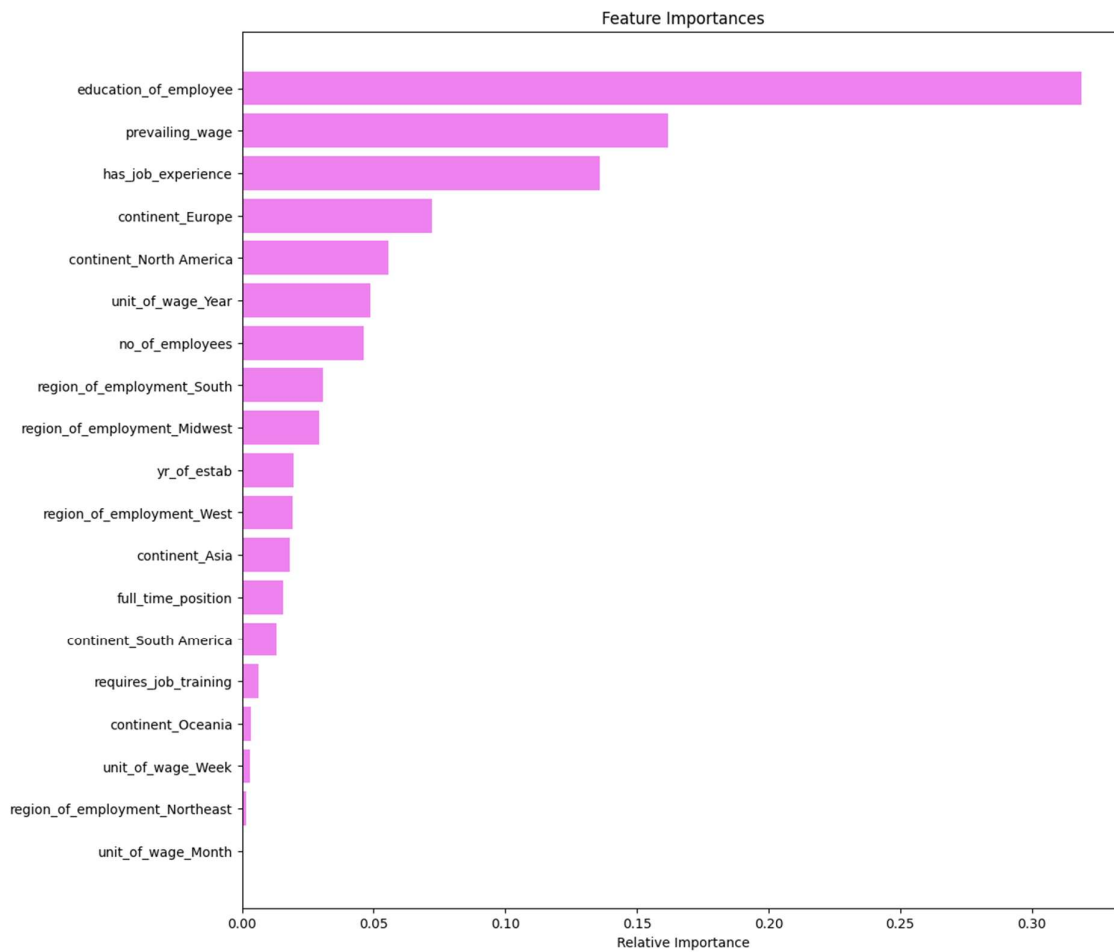


Fig 11.6 – Feature importance

- We can see that , `education\_of\_employee` ,
`has\_job\_experience` , `prevailing\_wage` and are the most important
features in predicting whether the visa will be certified or denied.

12. Actionable Insights & Recommendations

Based on our comprehensive analysis and the selected Gradient Boosting model tuned on original data, here are key insights and practical recommendations for OFLC to facilitate the visa approval process and the drivers that significantly influences the case status of visa being Certified or Denied.

12.1 Key Insights from Data Analysis and Model

1. Education Level is the Strongest Predictor

The education level of the employee (education_of_employee) emerges as the most influential factor in visa certification decisions. Higher educational attainment is strongly associated with a greater likelihood of certification, indicating that skill level and qualifications play a central role in approval outcomes.

2. Prior Work Experience Significantly Improves Certification Probability

Applicants with prior job experience (has_job_experience) are substantially more likely to receive certification. This suggests that demonstrated professional experience acts as an important credibility signal for visa approval.

3. Wage Level and Wage Structure Matter

Higher prevailing wages (prevailing_wage) are positively associated with certification outcomes, reflecting the preference for higher-value or specialized roles. Additionally, jobs offering annual wage structures (unit_of_wage_Year) show stronger certification likelihood compared to other wage units, indicating stability and long-term employment intent.

4. Geographic Factors Have Limited Influence

Regional and continental features (e.g., continent, region_of_employment) contribute relatively little to the model's predictions. This suggests that applicant qualifications and

compensation characteristics outweigh geographic considerations in certification decisions.

5. **Model Emphasizes Recall-Oriented Decision Making**

Confusion matrix analysis across training, validation, and test sets reveals consistently high recall for certified cases, with a higher rate of false positives. This reflects a deliberate trade-off favoring the identification of eligible applicants over strict rejection of borderline cases.

12.2 Recommendations

- **Prioritize Highly Educated and Experienced Applicants**

Given the strong influence of education and work experience, visa screening processes should prioritize candidates with higher educational qualifications and documented professional experience, particularly for roles requiring specialized skills.

- **Align Wage Offerings with Certification Expectations**

Employers should ensure that wage offers are competitive and aligned with prevailing wage standards. Clearly structured annual salary offers may improve certification likelihood by signalling job stability and long-term employment intent.

- **Support Employers in Role Structuring**

Employers may benefit from guidance on structuring job roles and compensation packages in ways that align with historical certification patterns, particularly for applicants with strong educational backgrounds.

- **Focus Less on Geographic Bias**

Since geographic factors show limited predictive importance, decision-making processes should continue emphasizing applicant merit rather than regional attributes.

- **Leverage Model for Early Risk Identification**

The recall-focused nature of the final model makes it suitable for early identification of likely certified applications. Borderline or high-risk

cases flagged by the model can be subjected to additional review rather than outright rejection.

- **Improve Data Collection for Future Analysis**

Collecting richer information on applicant skills, certifications, and role-specific requirements could further enhance model performance and decision transparency.

- **Adopt Threshold-Based Decision Support**

Instead of relying on a single classification threshold, decision-makers can adjust thresholds depending on policy objectives—prioritizing higher recall when the goal is inclusivity, or higher precision when stricter screening is required.

By leveraging data-driven insights from applicant qualifications, work experience, and wage characteristics, OFLC can improve decision consistency, reduce processing risk, and support fair, transparent, and skill-oriented visa approval policies.